OpenAffect-EEG: Auditing EEG Added Value Under Matched Label Resources

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Abstract

An EEG affect system's gain cannot be attributed to EEG when its comparator lacks the same calibration labels. OpenAffect-EEG makes the comparison executable by independently dosing participant and repeated-stimulus labels, fixing test support and population-training size, and pairing EEG predictors with same-resource EEG-free controls. Development on two cohorts shows how positive system-gain point estimates become uncertain or negative matched EEG increments. A design locked before FACED label access then evaluates 123 participants and 28 stimuli with complete resource grids, five training seeds, EEGNet and adapted LaBraM. Joint intervals include participant, stimulus, executed-seed and refit-rotation uncertainty. LaBraM's mean matched CCC increment is -0.0053 (95% interval $[-0.0112, 0.0009]$), within the pre-outcome ± 0.05 and ± 0.025 equivalence bounds. EEGNet remains inconclusive. Full-grid label permutations show no spurious-positive warning. Two locked sinusoidal positive controls failed; redesigned signed-power subsets recover large gains but remain post-failure exploratory diagnostics. These failures prevent a fully preregistered confirmation claim. Source-linked execution cases further distinguish valid partitions from supported deployment claims. The contribution is a resource-conditioned evaluation design and auditable evidence for model-specific attribution, rather than a universal claim that EEG lacks information.

1 Introduction

An affect-prediction system may receive not only EEG, but also ratings from the target user's earlier trials or from other people viewing the same stimulus. Such resources can be legitimate at deployment. The scientific question is whether the comparator receives them too: a favourable difference against an uncalibrated baseline measures the combined system, while the increment attributable to the evaluated EEG pipeline requires the same non-EEG labels on both sides. Repeated stimuli and individual self-reports make this distinction consequential in affective computing [8, 10, 16].

Trial-level leakage prevention is necessary but does not settle resource parity. A system can exclude every test trial yet receive target-user labels that its comparator lacks. Conversely, matching labels does not restore an unseen-user claim once calibration has exposed that user. We ask: *on fixed test trials, what EEG increment remains when both predictors receive the same participant and stimulus label resources?*

OpenAffect-EEG operationalizes this question as a two-axis resource-response surface. It changes one declared label budget at a time, preserves population-training row count, fixes the scoring support and pairs each EEG predictor with an EEG-free predictor using the same calibration rule. The score subtraction is elementary; the contribution is the design that makes its operands comparable and exposes when a deployment claim exceeds the recorded execution. It is complementary to diagnostic benchmarks that separate aggregate performance into controlled capabilities [33].

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The evidence has distinct roles. EmoEEG-MC and Urban Appraisal are development cohorts. Five-seed tail-adapted LaBraM adds stronger same-data sensitivity evidence. FACED supplies a previously unused cohort under a prospectively locked design, with refitted rotations, joint uncertainty and fixed equivalence margins. Its locked positive-control failures are retained, so the later successful diagnostic subsets do not become confirmatory evidence. The result is an evaluation boundary: under one declared estimand, adapted LaBraM's average FACED increment meets the equivalence bounds, while EEGNet remains uncertain. Neither result establishes universal EEG uselessness.

We contribute:

1. **A resource-conditioned evaluation design.** A fixed-support, participant–stimulus label-dose surface and constant-size population fits extend participant-only controls into a paired budget-dependent estimand.
2. **Multi-cohort evidence with explicit inferential boundaries.** Complete grids, adapted neural models, executed training seeds and refit rotations distinguish system gains, matched increments and average equivalence; negative and failed control results remain part of the evidence.
3. **Auditable consequences for real evaluation workflows.** Existing score contrasts change their attribution under matched labels, while source-linked execution cases distinguish valid trial splits, matched resources and supported deployment claims. Reusable contracts and aggregate evidence make these judgements inspectable without restricted EEG access.

2 Relation to Existing Evaluation Work

EEGain standardizes datasets, preprocessing, models, splits, metrics, and prediction logging [17]. FMScope audits frozen representations with subject-axis, spectral, layer, and within-subject diagnostics [22]. Calibration, joint person–stimulus holdout, and EEG toolboxes are also established [4, 5, 21, 24]. Most closely, Pandilova et al. [32] pair an EEG-free participant label-tendency baseline with calibrated and population qEEG comparisons on DEAP and DREAMER. They do not jointly dose target-participant calibration and other-participant ratings for test stimuli on a constant-size population fit. The participant-only baseline is an established comparator, not a novelty claim. Our additional unit of analysis is instead a *paired prediction contrast*: which labels each predictor received on a fixed test support, and what part of a system gain remains after matching those labels.

Identity probes, linear erasure, data cards, and provenance hashes are not claimed as individually novel [1, 12, 22, 30]. Outside EEG, diagnostic benchmarks separate aggregate scores into controlled capabilities, stress-test metrics on configurable known-truth data, and pair standard protocols with reusable code [25, 33, 34, 36]. OpenAffect-EEG adopts those benchmark qualities for a domain-specific estimand: resource-matched EEG added value in observed self-report prediction. Dose curves, simulation, and checklists support this design rather than providing independent novelty claims. The empirical distinction is that a matched participant-only contrast cannot reveal how the EEG increment changes after same-stimulus labels become shared resources (Section 5). We neither audit physiological mechanisms nor claim broader dataset coverage or higher classification accuracy than existing EEG suites.

Why the evaluation object transfers beyond EEG. The general issue is attribution under unequal resources. If a component-bearing system is scored as $S(1, R_1; T)$ and its comparator as $S(0, R_0; T)$, their difference combines the component, the change from R_0 to R_1, and their interaction. A component-specific increment instead compares $S(1, R; T)$ with $S(0, R; T)$ on fixed support T. Related evidence shows why this distinction is material: higher multimodal performance need not identify cross-modal interaction [14]; controlled vision–language comparisons attribute many reported differences to training data and hyperparameters [7]; and active learning, domain adaptation, and personalized federated learning change with annotation or local data budgets [3, 13, 28]. The invariant depends on the claim: an active-learning study may treat selected labels as the intervention and hold annotation cost fixed, whereas a modality ablation that claims added information should share calibration labels. Our cross-domain contribution is therefore a reusable resource-contract test for component attribution. FACED validates its consequence for EEG; it does not estimate the prevalence of mismatch in other fields.

Table 1: Complementary evaluation objects. This is a scoped comparison of published purposes, not an exhaustive absence-of-feature claim.

Approach	Question addressed	Additional object evaluated here
Trial/identity holdout	Do forbidden trials or identities cross partitions?	Dose-controlled resources with a fixed test and population size.
EEGain [17]	How do models compare in a standardized pipeline?	Does an EEG gain persist against a comparator with the same calibration labels?
FMScope [22]	What identity structure is encoded, and how does erasure affect decoding?	What labelled resources supported this prediction contrast and deployment claim?
Matched personalization [32]	How much do participant identity and target-user data explain qEEG performance?	A two-axis participant–stimulus label-resource surface with constant population size.
OpenAffect-EEG	What can be attributed to EEG under a declared resource contract?	Paired score contrasts plus verifiable resource and execution records.

3 Resource-Matched Evaluation

3.1 Trial graph and resource contract

Each trial has a participant identity i, stimulus identity j, observed report y, context, and source lineage. Windows inherit their complete trial's partition. Let T_b be the fixed test trials in identity block b and $D_b^{p,s}$ the population development data at doses (p, s). Let $C_{i,b}^p$ contain p labelled calibration trials from target participant i, on stimuli outside the test stimulus set. Repeated-stimulus dose s counts population ratings per test stimulus from other participants. We record a contract

$$\mathcal{R}_b(p, s) = \big(T_b, D_b^{p,s}, \{C_{i,b}^p\}_i, N_b, A_b, F_b, V_b\big), \tag{1}$$

where N_b is the fixed population-training row count, A_b lists additional label resources such as normative scores, and F_b, V_b record preprocessing fit and model-selection trials. This notation describes actual supplied resources; it is not an identity-information measure.

Repeated-stimulus rows replace context- and target-matched training donors, so population row counts remain fixed. Calibration rows do not fit the population model; they estimate a post-fit participant mean residual offset. Dose sets are deterministic and nested. Removing donors changes training composition; we control count and approximate label/context balance, not every property of the training distribution. Nor is total labelled-data budget fixed across the participant axis: additional calibration is the resource being measured.

The principal invariants are: no test trial in fit, selection, or calibration; identical test trials across cells and models; exact realised doses; no target participant in population fitting; and identical non-EEG labels for paired predictors. Test identities are initially jointly held out, but at nonzero doses the corresponding identities have been exposed. Released normative scores, where available, remain separately declared at zero stimulus-label dose. Zero dose therefore does not generally mean no stimulus knowledge. Unverifiable session, device, or upstream pretraining identities cannot support a strict isolation certificate.

3.2 Predictors and the estimand

The population prior P uses fit-set stimulus means, falling back to declared source normative ratings or the fit-global mean. PC adds the target user's mean calibration residual without EEG. E adds an EEG-predicted population residual to P, and EC calibrates E using exactly the same trial labels as PC. Both calibrated predictors use the same mean-offset rule. The residual is arithmetic relative to the specified prior, not an identified private emotion. Direct EEG prediction is retained as a secondary control.

For score functional Q on the same T_b, define

$$G_{\text{total}}(p, s) = Q(EC) - Q(P), \tag{2}$$

$$G_{\text{cal}}(p, s) = Q(PC) - Q(P), \tag{3}$$

$$\Delta_{\text{EEG}}(p, s) = Q(EC) - Q(PC). \tag{4}$$

Thus $G_{\text{total}} = G_{\text{cal}} + \Delta_{\text{EEG}}$ exactly. This is a telescoping score identity, not an additive decomposition of CCC into neural mechanisms. At a fixed s, both P and EC already receive the same repeated-stimulus labels: the unmatched resource in this particular contrast is participant calibration. Changing s studies the matched comparison under different shared stimulus resources, rather than estimating a pure causal effect of stimulus information. G_{total} can assess an entire system package; it cannot isolate EEG added value when calibration resources differ. At $p = 0$, $PC = P$ and the matched contrast reduces to $Q(E) - Q(P)$.

The common estimand is a uniform mean of Δ_{EEG} in CCC over the complete grid $\{0, 1, 2, 4, 8\}^2$. Development evidence averages five fixed blocks; FACED additionally averages both targets, executed seeds and fully refitted rotations under its locked design. We report paired MAE differences as a secondary metric; positive MAE differences mean greater error. Uniform grid weighting is a specified benchmark utility, not an estimated deployment-frequency distribution. Endpoint tables illustrate comparison choice but do not replace this primary summary. Every grid cell is retained in the aggregate release.

This estimand is specific to the fitted predictors and calibration rule. Equal label access does not imply equal ability to exploit those labels. Accordingly, a nonpositive increment diagnoses the evaluated pipeline, not the maximum information available in EEG. The uniform mean assigns equal weight to benchmark cells; it is not an optimal deployment policy. For an application, resource feasibility, participant time, recording cost, and application-specific cell weights must be specified separately.

3.3 Observable execution checks

The checker consumes a trial-identity table, an execution record, and boolean deployment requirements. Unlike a split-name lookup, it compares executed UID sets for population labels, calibration labels, test support, preprocessing, and model selection. Required missing evidence yields `unverifiable`, not a pass. A detected violation yields `block`; otherwise supplied observable evidence may `allow` the requested checks. Source identity authenticity and upstream pretraining exposure remain unverifiable even when observable checks pass.

For example, a calibrated target-user system can pass test-trial isolation yet fail a requested unseen-participant contract. Two predictors can also share every train/test split but fail resource parity because only one receives calibration labels. These are different checks with different inputs. A basic intersection checker is not expected to infer undeclared resource use. The resource checker complements it; it is not a universally more accurate detector.

4 Evidence Across Development and Held-Out Cohorts

4.1 Development evidence and stronger same-data models

The protocol was developed on EmoEEG-MC (30 participants, 42 stimuli; valence/arousal) and Urban Appraisal (62 participants, 56 images; valence) [38, 40]. The original complete resource grids cover band-power Ridge, frozen LaBraM and EEGNet under two bounded settings each. All 12 development-grid intervals include zero under the five-block fixed-fit analysis. These cohorts were already inspected, and Urban appraisal is not an identical emotion-induction replication. Their full grids, interval calibration and historical sensitivity checks remain in Appendix B.

To address the frozen-encoder limitation, existing v14 runs adapted LaBraM's last four transformer blocks, final normalization and regression head with five seeds. The mean matched increments are -0.0413 in EmoEEG-MC and -0.0418 in Urban (Table 2). These are negative same-data sensitivities with one fixed identity partition; they do not inherit FACED's prospective equivalence interpretation or become independent cohorts.

4.2 FACED: a locked design on a previously unused cohort

Before accessing FACED EEG samples or trial ratings, we locked source eligibility, resource doses, models, exclusions, estimands and equivalence margins. We used pinned NEMAR FACED v1.1.3 [8, 31]: 123 participants, 28 videos and 3,444 eligible trials, with no post-outcome exclusions. A header-only amendment harmonized two channel schemas to 30 scalp channels. Each trial uses the

last 30 seconds ending at video offset, common-average reference, a 0.5–45 Hz bandpass and 100 Hz resampling. Valence and arousal self-ratings are mapped from $[0, 7]$ to $[-1, 1]$.

Five participant folds crossed with four stimulus folds cover every eligible trial once per rotation. Five independently hashed rotations require full refitting, producing 100 splits; every split uses the complete $\{0, 1, 2, 4, 8\}^2$ resource grid. EEGNet uses validation-selected learning rate and training duration. LaBraM adapts its final four blocks, normalization and head. Both use the same five frozen training seeds and post-fit calibration rule. Test labels select neither checkpoints nor hyperparameters. Positive resource doses deliberately expose the corresponding identity, so these cells are resource-conditioned deployments rather than zero-shot identity tests.

The FACED primary statistic averages matched CCC differences over all resource cells and both targets. Joint intervals reuse the already executed predictions and share participant–stimulus multiplicities while resampling training seeds and complete refit rotations. This includes the executed training procedures, not uncertainty over arbitrary architectures or tuning policies. All observed analyses have 2,000 valid draws. The pre-outcome primary margin is ± 0.05 CCC, with ± 0.025 as a stricter sensitivity; equivalence requires the 90% decision interval to lie inside the margin. These are benchmark resolution choices, not clinically validated thresholds. A grid-average equivalence statement does not certify each target, participant or cell.

Adapted LaBraM's FACED mean is -0.0053 (95% interval $[-0.0112, +0.0009]$; 90% interval $[-0.0103, -0.0001]$), within both equivalence bounds. EEGNet's mean is -0.0414 (95% interval $[-0.0875, +0.0002]$); its 90% interval $[-0.0796, -0.0001]$ crosses the lower equivalence bound, leaving the decision inconclusive. The two models therefore support different levels of precision, not a universal absence of EEG information. These statistical decisions remain conditional on the declared design; the failed controls below limit a fully confirmatory reading.

The heatmap exposes a boundary hidden by the mean. At zero participant calibration, LaBraM's point increment changes from $+0.0170$ with no same-stimulus labels to -0.0212 with eight such labels. At the corresponding zero-stimulus corners, its increment is $+0.0170$ at participant dose zero and -0.0008 at dose eight. All four corners for both architectures are released, without choosing cells by sign. These are descriptive changes under different contracts, not a tested causal effect of label acquisition or proof that a particular cell has no EEG signal.

4.3 Negative controls, failures and post-failure recovery

Full-grid label permutations give mean increments -0.0400 (EEGNet) and -0.0058 (LaBraM). Neither triggers the prespecified spurious-positive warning; the similarity to observed estimates is descriptive, not a formal observed-versus-permuted test. Two completed EEGNet sinusoidal positive controls failed: raw-target encoding gave -0.1476, and residual-aligned encoding gave -0.0562, below the required $+0.10$ gate. Thus target alignment alone did not explain the failure. Both failures and their intervals remain in Table 4.

After observing those failures, signed-power controls separated positive and negative residual magnitudes across channels to avoid crop-dependent carrier phase ambiguity. A frozen, cost-controlled subset recovered $+0.6512$ with EEGNet and $+0.6303$ with LaBraM. It covers five selected splits, three stimulus doses and all participant doses and training seeds, not the full crossed test support. The earlier 52-fit incomplete full-grid attempt is retained and excluded from inference. These results show sensitivity to a redesigned, strong injected signal; they neither establish optimal use of real EEG nor retroactively repair the failed locked controls. FACED is an unused cohort with a prospectively locked observed-data design, but the complete pipeline does not meet a fully preregistered confirmation claim.

5 What the Evaluation Changes in Practice

Case 1: attributing an apparent system gain. At the predeclared maximum resource dose, the frozen LaBraM R1 system improves on the uncalibrated prior by $+0.0961$ CCC in EmoEEG-MC and $+0.0587$ in Urban. Supplying exactly the same calibration labels to the EEG-free comparator yields $+0.1420$ and $+0.0774$, leaving matched EEG differences -0.0459 and -0.0186. The original positive point estimate describes the system package; it does not identify added EEG value. Even the total-gain intervals include zero, so this is not a reversal of an established significant benefit.

Table 2: Existing five-seed adapted-model evidence. FACED intervals jointly resample participants, stimuli, executed training seeds and five fully refitted rotations; development-task intervals hold their single rotation fixed. Equivalence decisions apply to the uniform-grid mean and use the 90% interval. Failed locked controls limit the FACED confirmation claim.

Dataset	Model	ΔCCC	95% interval	Decision
FACED	EEGNet	-0.0414	$[-0.0875, +0.0002]$	uncertain
FACED	LaBraM tail	-0.0053	$[-0.0112, +0.0009]$	equivalent*
EmoEEG-MC	LaBraM tail	-0.0413	$[-0.0690, -0.0112]$	negative
Urban	LaBraM tail	-0.0418	$[-0.0755, -0.0064]$	negative

*FACED LaBraM also meets the narrower ± 0.025 margin. Development-task margins were not prespecified and are not used for a confirmatory equivalence claim.

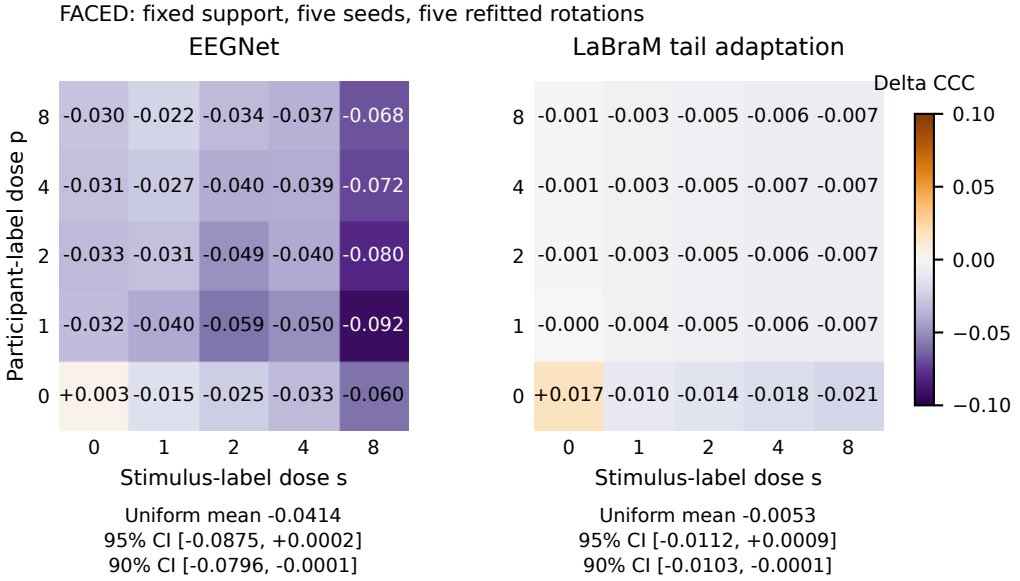


Figure 1: The complete FACED matched-increment grids. Positive values favor EEG; the shared scale and printed values retain dose heterogeneity. Displayed zeros are rounded, not exact equality. Each cell averages paired differences over executed seeds, rotations and targets on identical test support. Intervals below each grid describe the uniform mean. Pointwise and simultaneous cell bands are released separately; the heatmap does not establish cellwise equivalence. Negative and failed positive controls are reported in Table 4.

Appendix B reports all settings and the descriptive endpoint ranges. The practical correction is to change the comparator and attribution, not to claim that a deployed sensor is unnecessary.

Case 2: distinguishing calibration regimes. A single zero-resource comparison cannot answer what EEG adds after repeated stimulus labels become available. FACED's LaBraM surface ranges from positive to negative point increments across those shared budgets, while its complete uniform mean meets the fixed equivalence margins. Reporting the grid makes the budget assumption visible and prevents promoting the mean to an equivalence certificate for a selected deployment cell. This is the empirical consequence of adding a stimulus axis to a participant-only baseline.

Case 3: narrowing an external execution claim. For the same pinned EEGain split function and eligible trial universe, the checker blocks all five requests that jointly require held-out test participants and training–validation trial isolation. The same five traces allow the narrower held-out-test-participant claim. Generated marker windows carry lineage only; no external emotion classifier was retrained. This source-linked case changes a checkable claim about executed partitions, not

Table 3: Existing maximum-dose LaBraM R1 cases: point CCC contrasts. Changing the comparator changes the attribution. None of the total or matched endpoint ranges establishes a positive gain; full conditional ranges and other settings are retained in the development appendix.

Task	System $EC - P$	Calibration $PC - P$	Matched EEG $EC - PC$
EmoEEG-MC	$+0.0961$	$+0.1420$	-0.0459
Urban	$+0.0587$	$+0.0774$	-0.0186

the validity of EEGain's published model scores. Separate synthetic faults test resource parity and missing provenance; they are not attributed to external authors. The reusable interface accepts trial identities, executed resource records and explicit requirements, and returns `allow`, `block` or `unverifiable` with failed checks.

6 Discussion, Limitations and Artifacts

What is new, and what is not. Resource parity and identity-only baselines are established ideas. The proposed object combines participant and stimulus label doses, fixed test support, controlled population size and paired predictor outputs. Its value is the additional conclusion it permits: whether a claimed component increment persists under the stated calibration regime. It does not estimate a causal neural mechanism, optimize an adaptation policy, or require every resource in every benchmark to be identical. The matching contract must follow the component named in the scientific claim.

Field relevance and the audit's limits. Two independently locked model-assisted coding passes found no eligible learned no-EEG comparator in 24/24 publicly accessible EEG-affect papers. This supports an exploratory observation about missing comparisons in that cohort, not a claim that those papers used mismatched comparisons or reached invalid conclusions. Access conditioning, search wording and correlated model errors limit generalization; protocol unitization differs in four papers. The source codes, disagreement records and post-lock adjudication are released (Appendix A). These automated passes do not satisfy a two-human coding claim.

Limits of the scientific inference. Equal access does not ensure that the selected EEG architecture exploits the resources optimally. Failed locked controls and post-failure signal redesign remain material limitations even though later injected-signal recovery is large. FACED intervals include the executed seed and rotation choices, not all possible tuning or pretraining uncertainty. Older development intervals condition on fitted models. Pre-outcome margins define a benchmark resolution, not a clinically justified threshold; grid averaging can hide a useful cell, target or participant subgroup. Repeated-stimulus budgets, label quality and acquisition conditions differ between deployments. We establish neither an economic advantage of avoiding EEG acquisition nor broad performance outside the evaluated affect tasks. Cross-domain evidence motivates the transferable contract, rather than validating mismatch prevalence in other fields.

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OpenAffect-EEG makes a system gain and an EEG increment different reportable objects. Its strongest supported conclusion is specific: the comparator's label resources change what the evaluated score contrast warrants, and adapted LaBraM's average FACED increment meets the declared equivalence bounds under a pipeline with disclosed control limitations. Uncertain and adverse results remain visible instead of being converted into a universal verdict about EEG.

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A V14 Evidence, Frozen Analysis and Reproduction Boundaries

This appendix is the current v14 evidence layer. The following development appendices preserve earlier analyses on their original supports. Their use of "primary" refers to their own frozen development plan, not to the prospective FACED design. All v14 tables here are rebuilt from archived aggregate files; the manuscript integration reruns no training, bootstrap or simulation.

A.1 Pinned FACED acquisition and assignment

The source is FACED on NEMAR, nm000112 v1.1.3, under CC BY 4.0 [8, 31]. The protocol was locked on 14 September 2026 before EEG samples or participant rating values were read. The cohort contains 123 complete participants and 28 videos; all 3,444 trials pass the complete-support rule without post-outcome exclusions. Eligibility requires readable BDF, exactly one valid presentation of every video, at least 30 seconds ending at video offset, finite ratings and finite selected EEG samples. No rating is imputed and no outcome-dependent artefact rejection is performed.

Header-only preflight found 61 legacy and 62 modern channel schemas and sampling rates of 1000 Hz (68 files) or 250 Hz (55 files). The pre-outcome amendment renames legacy temporal channels and uses their common 30-channel scalp intersection, excluding A1/A2 or horizontal EOG channels. Video end is mapped to its nearest sample, then exactly 30 seconds are taken backwards. After conversion to microvolts, common-average reference and a fourth-order zero-phase 0.5–45 Hz Butterworth bandpass use two-second reflections of the within-video window. Polyphase resampling yields 30 by 3,000 samples. No post-video rating-period signal enters the padding. Ratings use $(y - 3.5)/3.5$. The Ridge view is per-channel log10 Welch power in five fixed bands with a two-second window; it is a completed sanity baseline, not a new neural model or a separately claimed confirmation result.

All 2,500 already executed Ridge cell records are retained as scalar metric rows with source-record hashes. Their descriptive uniform mean is -0.0020 CCC (valence -0.0052, arousal $+0.0012$). These are means of the existing cell scores; no new prediction or bootstrap was run. No joint Ridge interval was archived, so we do not assign a significance or equivalence decision to these point summaries.

Five participant folds and four stimulus folds give 20 test cross-products per rotation. The five rotation seeds are 20260914, 20260931, 20260948, 20260965 and 20260982; the global split index runs from zero to 99. Population fitting excludes test participants and initially excludes test stimuli; nonzero stimulus doses add declared ratings from other participants by replacing donor rows. Participant calibration remains outside population fitting. Each rotation covers every eligible trial once, but rotations and seeds are not additional independent participants. The recorded 17,220 rows per full-grid cell count repeated trial occurrences across rotations; there are still only 3,444 distinct eligible trials and 123 independent participant identities.

A.2 Neural training and uncertainty

EEGNet uses learning rates 10^{-3} and 10^{-4}, dropout 0.25, AdamW weight decay 10^{-4}, batch size 128, at most 100 epochs and patience 15. LaBraM adapts the final four transformer blocks, final normalization and a linear regression head, with backbone/head learning rates $10^{-5}/10^{-3}$, AdamW weight decay 0.01, batch size eight, at most 40 epochs and patience eight. Training uses random ten-second crops; evaluation is deterministic. Validation MAE selects duration, followed by full refitting on permitted training plus validation trials. Training seeds are 2026091401 through 2026091405. Participant-dose changes use post-fit calibration and do not cause redundant EEG retraining. Checkpoints, pretraining exposure and arbitrary hyperparameter search policies remain outside the observable claim contract.

The analysis computes a CCC per target on each scoring split, pairs predictors before averaging, and shares participant–stimulus multiplicities across compared predictors and resource cells. Executed

seeds and fully refitted rotations are resampled within the frozen numerical procedure. Percentile 95% estimation intervals and 90% TOST-compatible decision intervals use 2,000 requested draws; all full-grid stages have 2,000 valid draws and each signed-power subset has 1,999. Cell CSVs additionally report MAE differences, pointwise intervals and a max-absolute-deviation simultaneous band. This bootstrap's coverage is not independently established by the older Gaussian five-block simulations: those validate the older fixed-fit development statistic only. Five seeds and five rotations also do not cover every possible model, tuning strategy or acquisition domain.

The primary equivalence margin ± 0.05 and strict sensitivity margin ± 0.025 were fixed before FACED label access. They operationalize two absolute CCC resolutions, not a patient-, user- or task-validated minimum important effect. We report the continuous estimate and both decision intervals so readers can apply another justified bound. Equivalence is limited to the average of the declared cells and targets. Older development datasets have no prospective equivalence claim, and no FACED per-target or subgroup equivalence is inferred from the pooled mean.

Table 4: All completed FACED neural controls, including failures. Signed-power rows use an exploratory subset with 85 participants, 23 observed stimulus identities, 15 resource cells and five sampled split indices; they do not replace either failed full-grid control.

Model	Control	ΔCCC	95% interval	Gate
EEGNet	label permutation	-0.0400	$[-0.0856, +0.0002]$	pass
LaBraM tail	label permutation	-0.0058	$[-0.0077, -0.0041]$	pass
EEGNet	raw-target sinusoid	-0.1476	$[-0.2398, +0.0573]$	warning
EEGNet	residual sinusoid	-0.0562	$[-0.1193, +0.0267]$	warning
EEGNet	signed-power subset	$+0.6512$	$[+0.5968, +0.7854]$	pass
LaBraM tail	signed-power subset	$+0.6303$	$[+0.5770, +0.7595]$	pass

A.3 Complete control chronology and source integrity

The original raw-outcome sinusoid failed the EEGNet $+0.10$ recovery gate. Because the model learns an outcome-minus-prior residual and adds it to the prior, raw-target encoding could double-count that prior. The dated 17 September amendment encoded the residual, but the second completed control also failed. A deterministic oracle residual attained $+0.6176$, showing the gate is attainable under the scoring code, not proving neural learnability. A subsequent single-cell signed-power pilot separated residual signs across channels, reducing carrier phase ambiguity under random crops. This is a post-failure mechanism hypothesis and diagnostic, not a preregistered test.

The signed-power full-grid attempt was stopped after 52 of 5,000 EEGNet candidate fits because of measured cost; its partial record is excluded from inference. Before the subset restart, split indices $0, 20, 40, 60, 80$, stimulus doses $0, 2, 8$, all five participant doses and all five training seeds were fixed. This yields 150 EEGNet candidates (75 selected) and 75 LaBraM fits. Its realized support includes 85 participants and 23 videos. Both models recover the redesigned signal, with the intervals in Table 4; neither replaces a failed full-grid control. The LaBraM full-grid permutation has 2,500 fits and no spurious-positive warning. Completed controls, failed controls, the stopped attempt and their different supports are disclosed without changing observed predictions or original margins.

Ten completed neural analysis sets are released as immutable summary JSONs and cell CSVs with source SHA-256 records. The paper builder checks every source hash, complete cell key set, primary mean and equivalence decision before formatting tables. Original machine fields labelled `confirmatory` record the invocation scope; the paper's interpretation explicitly withholds fully preregistered confirmation because the locked controls failed. A resource-corner table retains every corner for both FACED architectures; neither signs nor significance select the displayed cases. Ridge completion, ingestion, structural-QC and assignment summaries are also released. Trial-level predictions, participant identifiers, tensors and weights remain excluded.

A.4 Model-assisted literature coding and independent execution

The literature sampling frame and coding manual were frozen before coding. The original main/reserve cohort suffered access attrition; a dated amendment continued the same hash order to form a 24-paper public-full-text sensitivity cohort, four papers per year in 2021–2026. Two separately

instructed model operators locked their codes before comparison. Both found no eligible learned no-EEG comparator in 24/24 papers. Raw agreement is 100% for comparator presence and parity, but Cohen's kappa is undefined for a constant category. Claim type agreed in 23/24 papers; confidence agreed in 20/24. The operators created 60 versus 57 protocol units; post-lock adjudication resolved omissions and material splits for three papers, while one target-granularity choice remains unresolved. No protocol-level kappa is inferred from unaligned units. The coders share a model family and may share errors. This does not meet the original two-human condition or estimate representative mismatch prevalence; absence of a comparator is not evidence of an invalid comparison.

The separately instructed artifact operator used a clean Python environment. The initial public quickstart exposed a stale release manifest; the controlled builder and synchronized manifest repaired that failure. The final checks verify executable software and aggregate evidence reconstruction, not full raw-to-paper EEG retraining. The acceptance type remains `author_assisted`, with no independent-human claim. Model assistance also supported coding, software maintenance and manuscript drafting; authors are responsible for source checks and all scientific interpretations.

A.5 Reproduction modes and computational boundary

The reviewer quickstart validates the release, executes a synthetic contract example, rebuilds earlier source-linked cases and rebuilds all v14 result tables from the public aggregate archive. The following v14 command performs only deterministic verification and formatting:

```
python scripts/build_v14_paper_assets.py --verify \
  --output /tmp/openaffect-v14-table-check
```

Full scientific reproduction is a separate, explicit data-owner workflow: obtain pinned source files, run the released signal/QC and assignment CLIs, execute EEGNet or LaBraM with the frozen configurations, and run the joint analysis on locally retained predictions. The command map, including all failed controls and the fixed diagnostic subset, is in `docs/v14_reproduction.md`. The manuscript integration itself repeats none of those experiments. Exact wall-clock and energy totals were not logged consistently across all earlier runs, so no complete training-cost claim is made. A remaining scientific limitation is that strong synthetic recovery cannot establish sensitivity to all realistic EEG effects.

B Preserved Development Evidence and Diagnostics

These are the pre-v14 development analyses on their original supports. Fixed-fit intervals and their simulation validation do not validate the FACED joint bootstrap.

B.1 Evaluation coverage

We evaluate EmoEEG-MC (30 eligible participants, 42 stimuli; valence and arousal) and Urban Appraisal (62 participants, 56 street images; genuine valence trials only) [38, 40]. Deterministic identity blocks test every eligible participant and stimulus once, yielding 252 and 695 fixed test trials. The design tests five diagonal blocks, not every possible participant–stimulus pair. Urban is an external appraisal task, not an identical replication of emotion induction, and neither dataset is an untouched confirmatory cohort.

Each task includes band-power Ridge, frozen LaBraM, and standard Braindecode EEGNet [15, 18]. Two bounded settings per model, five identity blocks, and all 25 dose cells yield 1,500 model-cell results; these cells are not independent observations. Ridge settings vary the candidate regularization grid, while EEGNet settings vary learning rate. Validation selects hyperparameters and training duration. Population refitting uses training and validation trials but never test or calibration trials. We record 200 nonconstant EEGNet selection/refit histories as execution evidence, not as proof of optimization. The Appendix reports full preprocessing, coverage, budgets, and historical model panels.

The archived records show that, among 50 direct and 50 residual EEGNet fits per task, 12/10 Emo and 8/7 Urban fits select epoch one; median selected epochs are 18/17.5 and 11/7.5. Training objectives decrease in the median for all four groups. This rules out every network remaining at its initial fit, but does not establish optimal tuning or a positive held-out EEG signal.

B.2 A favourable total gain need not be an EEG gain

Table 5 applies the same predeclared maximum dose to every setting. For EmoEEG-MC LaBraM R1, EC improves on the uncalibrated prior P by $+0.0961$. Yet the same calibration labels improve the EEG-free predictor by $+0.1420$, leaving a matched EEG difference of -0.0459 $[-0.1098, +0.0139]$. In Urban, the corresponding quantities are $+0.0587$, $+0.0774$, and -0.0186 $[-0.0563, +0.0126]$. The system-level point gains are positive; the EEG-specific contrasts are not established as positive. This is the judgement that resource matching changes.

The primary full-grid summaries in Table 6 tell the same bounded story: all 12 task–setting block-t intervals include zero. The conclusion is unchanged under released percentile, basic, and earlier t-normal sensitivities. No equivalence margin was prespecified, so this does not establish equivalence, universal reversal, absence of EEG information, or a ceiling on future EEG methods. The secondary MAE contrast for LaBraM R1 indicates increased error in both tasks under the primary construction; other configurations are uncertain or interval-sensitive. Appendix C reports all CCC and MAE settings rather than selected rows.

Post-primary checks preserve that interpretation. An alternative identity partition and complete refit shifts any point estimate by at most 0.0135 (Emo) or 0.0154 (Urban), with all 12 intervals covering zero. Across three EEGNet initialization schedules at the four resource corners, all 12 task–setting–run intervals cover zero and the largest point span is 0.0097. Last-block LaBraM residual fine-tuning gives -0.0110 $[-0.1041, +0.0820]$ and -0.0404 $[-0.1159, +0.0351]$. These are bounded sensitivities run after the primary analysis, not independent cohorts or an unconditional retraining interval.

Figure S1 exposes heterogeneity hidden by the uniform mean. For example, Urban EEGNet L2 has positive point estimates in every cell, ranging from $+0.0003$ to $+0.0177$; these do not establish positive increments at the reported precision. None of the 300 cells has a positive pointwise CCC range. Two Urban LaBraM R1 cells have negative pointwise ranges, but neither survives its within-model dose-family band. These ranges are descriptive because the cell estimands failed full calibration. We retain all 48 task–setting–corner comparisons $(0, 0), (8, 0), (0, 8), (8, 8)$ in a separate table; no cell selection or averaging of interval endpoints is used.

Why is the matched comparator material? From zero dose, eight target-user calibration trials increase EEG-free PC CCC by $+0.2036$ in EmoEEG-MC and $+0.1564$ in Urban. Eight other-participant ratings per test stimulus increase it by $+0.4907$ and $+0.3408$, respectively; both interval constructions are positive. The fact that labels help is unsurprising. Their magnitude relative to the matched EEG contrasts shows why unequal label access can change the attribution attached to a system score.

B.3 Uncertainty and evaluator validity

The crossed bootstrap applies identical participant and stimulus multiplicities to every paired predictor, cell, and block. Of 2,000 draws, 1,985 are valid for EmoEEG-MC and 2,000 for Urban. The primary two-sided 95% interval uses the crossed-bootstrap standard error and a t_4 critical value because the final estimand averages five predeclared identity blocks. Percentile, basic, and the earlier within-block t-normal constructions are sensitivity checks. Every construction conditions on fitted predictions and omits retraining uncertainty. Pointwise ranges and separately released dose-family bands do not support calibrated cell selection or all-model multiple comparisons.

The relevant cluster sizes are those within each scoring block: Emo has six participants and eight or nine stimuli per block; Urban has 12 or 13 participants and 11 or 12 stimuli. The whole-dataset participant count does not by itself validate a mean of small-block CCCs. We therefore distinguish the earlier single-block simulations below from the topology-matched five-block check in Appendix E.

The new topology-matched validation executes 300 replicates for each of four Gaussian scenarios per task, with 2,000 bootstrap draws and no failed runs. For the primary uniform mean, block-t coverage is 0.993 for Emo and 1.000 for Urban under trial signal, and 0.980 and 0.983 under participant-constant signal; the two non-signal checks cover at 1.000. The corresponding Wilson Monte Carlo lower bounds are 0.957–0.987. The minimum cellwise coverage is still 0.917, so this validation supports only the predeclared uniform mean. The block-t correction was introduced after the earlier interval diagnostics failed, but its critical value follows the five-block aggregation and does not use the observed EEG effect signs or magnitudes. Its evidence remains limited to these four Gaussian

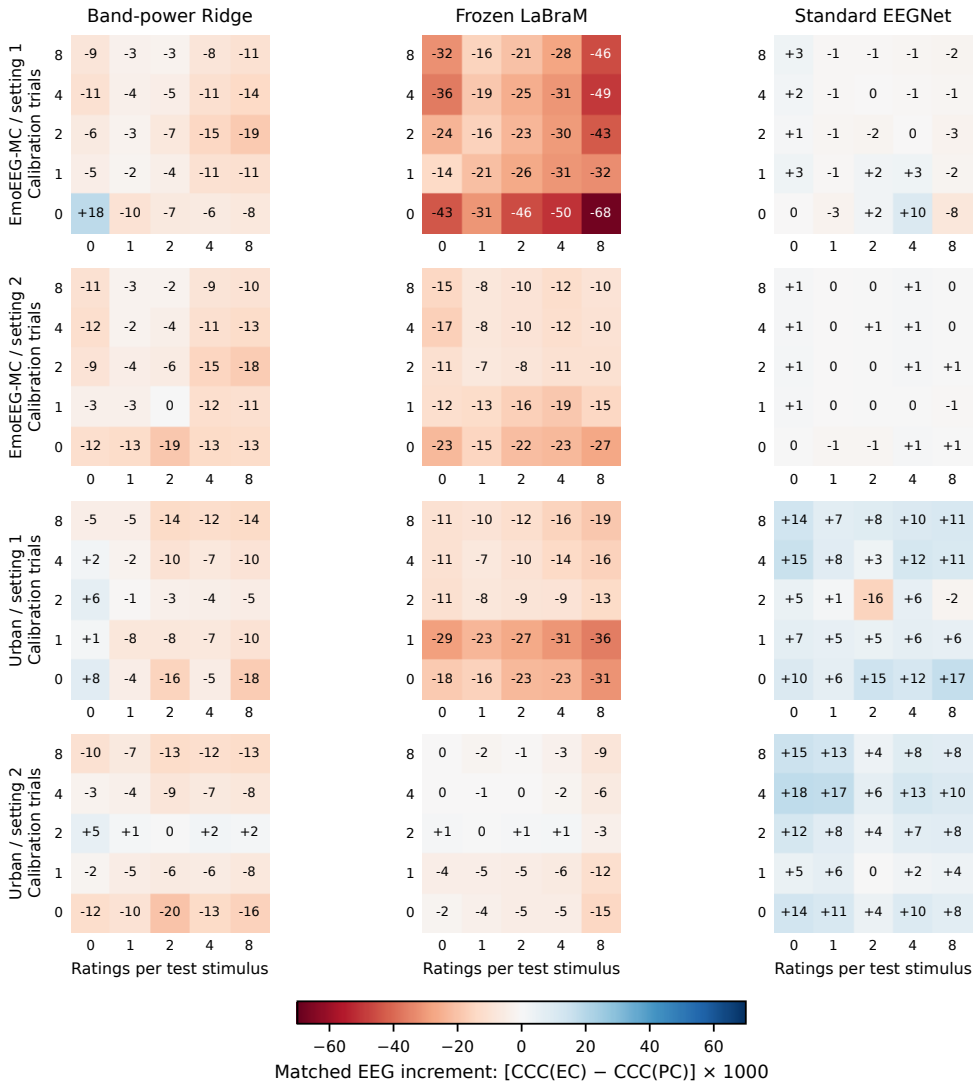


Figure S1: All 12 matched-increment grids, with every dose retained. Entries and color scale are $1000\Delta_{\mathrm{EEG}}$; rounded zero is not exact equality. Rows distinguish tasks and settings (R1/L1, R2/L2); columns distinguish model families. Positive values favor EEG. Each value averages five block-level paired CCC contrasts. This is a descriptive display of already inspected data, not an additional confirmatory test. Conditional pointwise stability ranges and within-model dose-family bands accompany every cell in the released CSV.

Table 5: Changing the comparison at the predeclared maximum dose $(p, s) = (8, 8)$. Total system gain $EC - P$ includes the EEG-free calibration gain $PC - P$. The last column is the paired percentile interval for $EC - PC$, not a difference of interval endpoints. These descriptive endpoints do not replace the primary uniform-grid mean.

Task	Setting	$EC - P$	$PC - P$	$EC - PC$	Paired interval
EmoEEG-MC	Band-power R1	+0.1312	+0.1420	-0.0108	[-0.0429, +0.0219]
EmoEEG-MC	Band-power R2	+0.1318	+0.1420	-0.0102	[-0.0403, +0.0206]
EmoEEG-MC	LaBraM R1	+0.0961	+0.1420	-0.0459	[-0.1098, +0.0139]
EmoEEG-MC	LaBraM R2	+0.1325	+0.1420	-0.0095	[-0.0419, +0.0243]
EmoEEG-MC	EEGNet L1	+0.1400	+0.1420	-0.0020	[-0.0065, +0.0027]
EmoEEG-MC	EEGNet L2	+0.1420	+0.1420	0.0000	[-0.0040, +0.0043]
Urban	Band-power R1	+0.0639	+0.0774	-0.0135	[-0.0561, +0.0362]
Urban	Band-power R2	+0.0643	+0.0774	-0.0131	[-0.0398, +0.0199]
Urban	LaBraM R1	+0.0587	+0.0774	-0.0186	[-0.0563, +0.0126]
Urban	LaBraM R2	+0.0689	+0.0774	-0.0085	[-0.0373, +0.0172]
Urban	EEGNet L1	+0.0885	+0.0774	+0.0111	[-0.0147, +0.0360]
Urban	EEGNet L2	+0.0851	+0.0774	+0.0078	[-0.0064, +0.0252]

Table 6: Primary uniform-grid mean matched EEG increment in CCC, with two-sided 95% five-block t intervals (df=4). Each task averages five fixed identity blocks and all 25 dose cells. Settings are sensitivity checks, not independent replications.

Setting	EmoEEG-MC: mean [interval]	Urban: mean [interval]
Band-power R1	-0.0069 [-0.0464, +0.0327]	-0.0060 [-0.0631, +0.0511]
Band-power R2	-0.0091 [-0.0409, +0.0226]	-0.0069 [-0.0444, +0.0307]
LaBraM R1	-0.0320 [-0.0883, +0.0242]	-0.0173 [-0.0607, +0.0261]
LaBraM R2	-0.0137 [-0.0492, +0.0219]	-0.0035 [-0.0312, +0.0242]
EEGNet L1	-0.0001 [-0.0065, +0.0064]	+0.0073 [-0.0117, +0.0264]
EEGNet L2	+0.0003 [-0.0046, +0.0052]	+0.0086 [-0.0094, +0.0265]

families. We therefore make no calibrated cellwise, equivalence, or unconditional retraining claim. Earlier single-block tests in Appendix C.5 cannot widen that scope.

B.4 Scope checks and retained negative evidence

Earlier EmoEEG-MC and MusicEEG surfaces, compact versus standard EEGNet, CBraMod, LaBraM random controls, frozen EMOD, and calibration variants remain in Appendix H.2. They use distinct supports or protocols and are not replications of the primary grid. The earlier MusicEEG LaBraM mean increment is 0.0182 $[-0.0161, 0.0452]$, an uncertain auditory result that we retain rather than omit. Resource-conditioned model selection likewise has no stable mean CCC benefit (Appendix H.1).

Identity probes and subspace perturbations are representation diagnostics, not causal evidence. The ds006866 analysis concerns condition exposure because image identity is unavailable. DENS is missingness-limited; SEED-V is an authorized aggregate supplement, not an openly redistributed benchmark. The repaired EMOD single-seed check is neither a full original-task reproduction nor evidence against EMOD, and mdJPT has no completed numerical audit here. These records bound model and domain coverage; none is promoted to primary evidence.

C Development Resource-Matched Coverage Analysis

This section and the preserved development tables use the frozen `final_closure_v9` evidence. The historical supplement preserves the original experiments, figures, and their support. They are not additional independent replications of this panel, and their historical use of "primary" denotes their own frozen analysis plan rather than replacing the present main summary.

C.1 Source and coverage details

EmoEEG-MC uses snapshot 1.0.7; Urban Appraisal uses snapshot 1.0.0 [38, 40]. Emo has 1,260 eligible trials from 30 participants and 42 stimulus identities. Urban uses 3,471 valence trials from 62 participants and 56 images; other scales are not copied into valence/arousal labels. A source recording with confirmed truncation was excluded before prediction inspection, as documented in the frozen coverage manifest. Urban's dataset-description license says CC0, while its README says CC BY 4.0. We document both declarations and do not redistribute the source recording or image material.

SHA-256 ordering with design seed 20260908 assigns participant and stimulus identities to five blocks. Block b supplies test pairs; block $b + 1$ modulo five supplies validation pairs; population training initially uses identities from the other blocks. Test pairs lie on diagonal blocks only. Coverage means every eligible participant and stimulus is tested once, not every possible trial. The same test support is retained throughout the dose grid. Stimulus-label injection may use participants from the validation block; only the tested participants are categorically excluded from population fitting at all doses. This is not a claim of fully disjoint validation identities after injection.

C.2 Bounded model settings and preprocessing

Ridge R1 selects alpha from $\{0.1, 1, 10, 100\}$ and R2 from $\{1, 10, 100, 1000\}$. Both band-power and frozen LaBraM use the two grids. The historical Emo feature key `de` selects channel log-bandpower; it does not make this panel a separate differential-entropy implementation. Standard EEGNet is Braindecode 1.2.0 with F1=8, D=2, F2=16, dropout 0.25, AdamW weight decay 10^{-4}, batch size 128, at most 100 epochs and patience 15. L1 and L2 use learning rates 10^{-3} and 10^{-4}. Validation MAE selects training length; the selected length is refitted on train plus validation. Both direct and residual branches are retained. Calibration changes post-fit offsets only; it is not a sweep of few-shot fine-tuning algorithms.

Emo uses the existing 100-Hz raw trial archive and ten-second EEGNet crops. Urban uses 64 EEG channels in physical volts, excludes ECG and GSR, applies common-average reference and polyphase resampling from 500 to 100 Hz, and uses the three-second image interval before the rating response. At evaluation, window predictions are averaged within a trial. Downstream learned normalization is fitted without test or calibration trials. Frozen LaBraM uses the project's recorded checkpoint and channel/unit adaptation; its upstream training-data identity provenance is not certified here.

C.3 Complete primary and secondary summaries

All tables below are generated from the same hashed fitted predictions as the main tables. The V13 primary interval recomputes the crossed bootstrap with a five-block critical value; it performs no new fitting or model selection. The score identity $EC - P = (PC - P) + (EC - PC)$ is checked for all 300 task–setting–dose combinations after block aggregation, with maximum numerical error below 10^{-12}. This validates a numerical identity, not a causal decomposition. Paired intervals are copied from paired contrast outputs, never constructed by subtracting separate confidence bounds.

Table 7: EmoEEG-MC: uniform-grid EC-PC CCC. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	-0.0069	[-0.0333, +0.0221]	[-0.0435, +0.0298]
Band-power R2	-0.0091	[-0.0306, +0.0145]	[-0.0385, +0.0202]
LaBraM R1	-0.0320	[-0.0676, +0.0104]	[-0.0842, +0.0201]
LaBraM R2	-0.0137	[-0.0376, +0.0139]	[-0.0466, +0.0192]
EEGNet L1	-0.0001	[-0.0040, +0.0047]	[-0.0060, +0.0059]
EEGNet L2	+0.0003	[-0.0028, +0.0040]	[-0.0042, +0.0048]

Table 8: EmoEEG-MC: uniform-grid EC-PC MAE. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	+0.0047	[-0.0004, +0.0099]	[-0.0019, +0.0113]
Band-power R2	+0.0039	[0.0000, +0.0078]	[-0.0013, +0.0090]
LaBraM R1	+0.0151	[+0.0061, +0.0242]	[+0.0034, +0.0269]
LaBraM R2	+0.0065	[+0.0016, +0.0112]	[+0.0001, +0.0129]
EEGNet L1	+0.0002	[-0.0008, +0.0013]	[-0.0012, +0.0016]
EEGNet L2	+0.0002	[-0.0004, +0.0009]	[-0.0007, +0.0011]

Table 9: Urban: uniform-grid EC-PC CCC. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	-0.0060	[-0.0432, +0.0376]	[-0.0519, +0.0398]
Band-power R2	-0.0069	[-0.0312, +0.0216]	[-0.0370, +0.0233]
LaBraM R1	-0.0173	[-0.0482, +0.0112]	[-0.0522, +0.0175]
LaBraM R2	-0.0035	[-0.0238, +0.0150]	[-0.0257, +0.0187]
EEGNet L1	+0.0073	[-0.0056, +0.0217]	[-0.0080, +0.0226]
EEGNet L2	+0.0086	[-0.0043, +0.0215]	[-0.0059, +0.0230]

C.4 Resource effects and statistical limits

For the EEG-free prior, participant-dose changes from $(0, 0)$ to $(8, 0)$ are $+0.2036$ $[+0.0955, +0.3116]$ on Emo and $+0.1564$ $[+0.0492, +0.2637]$ on Urban. Stimulus-dose changes to $(0, 8)$ are $+0.4907$ $[+0.3581, +0.6233]$ and $+0.3408$ $[+0.2161, +0.4655]$. Joint changes to $(8, 8)$ are $+0.6327$ $[+0.4810, +0.7844]$ and $+0.4181$ $[+0.2928, +0.5435]$. These are heuristic t-normal sensitivity intervals; the corresponding percentile intervals also exclude zero. They are not the same contrasts as $PC - P$ at a fixed stimulus dose in the main endpoint table.

The bootstrap reuses crossed participant and stimulus multiplicities across all fitted predictors. Averages are computed within block and then across blocks rather than pretending repeated grid predictions are new observations. The finite-cluster coverage diagnostics and invalid-draw counts constrain interpretation. The primary block-t interval addresses the uniform-grid mean only and remains a fixed-fit uncertainty summary, not a repeated-training confidence statement or equivalence test. Neither isolated pointwise ranges nor dose-family bands imply simultaneous coverage over every method, task, and secondary endpoint inspected during benchmark development.

C.5 Known-truth evaluator stress tests

We tested the estimator on outcomes with known construction before interpreting the EEG results. In the v6 oracle suite, 16 settings with 100 replicates each produced the following operating characteristics. "Detection" denotes a strictly positive interval and is not a universal power guarantee.

Table 10: Urban: uniform-grid EC-PC MAE. Paired pointwise intervals condition on fixed fitted predictions; t-normal is a heuristic sensitivity analysis.

Setting	Mean	Percentile interval	t-normal interval
Band-power R1	+0.0063	[-0.0012, +0.0144]	[-0.0024, +0.0151]
Band-power R2	+0.0037	[-0.0009, +0.0082]	[-0.0015, +0.0089]
LaBraM R1	+0.0091	[+0.0036, +0.0148]	[+0.0027, +0.0156]
LaBraM R2	+0.0038	[+0.0003, +0.0073]	[-0.0002, +0.0078]
EEGNet L1	+0.0027	[-0.0009, +0.0068]	[-0.0016, +0.0071]
EEGNet L2	+0.0012	[-0.0013, +0.0041]	[-0.0019, +0.0043]

Generator	Recorded behavior	Interpretation boundary
Labels only	mean difference exactly zero	algebraic negative control only
Independent noise	0/16 cells detected	finite evidence, not a proof of zero false positives
Trial signal	detection 0.85–1.00; coverage 0.92–1.00	detects the injected trial signal in these settings
Participant constant	minimum coverage 0.82	exposes finite-cluster weakness; no biological or causal interpretation

The v7 finite-cluster study planned 9,600 fits; 9,584 were valid and 16 were rejected by the declared support rule. In the target 32×24 cluster setting, percentile, basic, and t-normal coverages were 0.933, 0.930, and 0.957. Their minima in the deliberately small 8×6 setting were 0.837, 0.820, and 0.940. The often wider t-normal interval is therefore retained as a sensitivity analysis, not promoted as a generally calibrated solution.

C.6 Rule basis and three-state decisions

The checker implements observable evidence rules rather than inferring scientific validity from a score. Every rule returns allow, block, or unverifiable; the last state is preserved when required provenance is absent.

Rule family	Required evidence	Example boundary
Trial lineage	trial UIDs in every partition	repeated derived windows block trial-isolation claims
Identity holdout	participant/stimulus UIDs	overlap blocks only the identity claim it violates
Preprocess/selection	fit and selection UIDs	missing provenance is unverifiable, not a pass
Matched resources	predictor-specific label access	unequal calibration blocks an EEG-added-value claim
Fixed support	test-set hashes by comparator	unequal test support blocks a paired comparison

These rules operationalize documented leakage and benchmark-reporting concerns [4, 6, 19, 34]; they do not certify data truth, upstream pretraining history, or causal validity.

D Source-Linked Execution Cases and Reuse

The executed EEGain commit is `d6892f5586181f345969651a9baedabd828bb1c5`; its `EEGDataloader.split_train_val` is in `eegain/data/loader.py`. LibEER is pinned to `dddff9776dbdae21195fe320dff0a5ba61628a18`; we execute `get_split_index`, `index_to_data`, and the empty-validation branch at lines 100–102 of `LibEER/EEGNet_train.py`. File hashes, selected source ranges, case origin, and outputs are recorded in `external_audit/public_summary.json` and its paired case table. Marker windows are generated from real trial identities, not independent EEG measurements. No external classifier score is inferred from the branch tests.

The same EEGain execution record produces different answers to different contracts: its held-out test participants remain excluded, while an additional requirement that validation trials be disjoint from training can be violated. Changing the claim to what was actually isolated is one valid repair. When a strict validation-trial contract is required, partitioning complete trials before generating windows is the appropriate construction instead. LibEER's explicit train/validation/test path passes the exercised split checks; its empty-validation fallback should not be treated as independent validation. The machine-readable repair record `submission_closure_v11/external_claim_repair.json` verifies this strict-block/scoped-allow pair across all five exercised EEGain folds while holding the pinned function, trial universe, and execution output fixed. The repair changes the supportable claim; it does not alter or re-estimate a model score.

```
python scripts/run_workflow_contract_toy.py toy_contract
python -m openaffect_eeg.workflow_audit \
  --trials toy_contract/trials.tsv \
  --execution toy_contract/unmatched_calibration_execution.json \
  --contract toy_contract/unmatched_calibration_contract.json \
  --output toy_contract/checked.json
```

The mismatch example exits with the documented blocked status; the matched example passes observable checks. Omitting preprocessing evidence preserves uncertainty. This is a developer-operated executable example, not independent human acceptance. The checker validates consistency of supplied execution records, not adversarially falsified provenance.

AI-assisted development. AI assistance was used for protocol and code drafting, debugging, literature triage, analysis support, and manuscript development. The recorded EEG predictions, bootstrap summaries, and contract decisions are generated by the documented numerical models and deterministic checks, not by an LLM acting as an emotion predictor or audit judge. Automated checks and agent-run examples are not independent human review. Authors remain responsible for source verification, scientific interpretation, and submission approval.

E Matched-Grid Display and Learning Diagnostics

E.1 Complete cell evidence and deployment interpretation

The display in Figure S1 is generated directly from `table_added_value_contrasts.csv`. It preserves all 300 task–setting–dose rows, with their paired pointwise CCC/MAE intervals and within-model dose-family CCC bands. The four resource corners give 48 comparisons across all settings, including unfavorable and uncertain results. The files `matched_increment_all_cells.csv` and `deployment_corners.csv` retain unrounded values. A common symmetric color scale is used, with entries multiplied by 1000 for legibility. The visualization was added after examining the results and is descriptive.

The four corners answer different deployment questions: no target-user calibration or repeated-stimulus ratings; calibration only; repeated-stimulus ratings only; and both. Released normative information remains separately declared. These scenarios do not make ratings costless or assert that stimuli will recur in a real application. A weighted application score requires weights selected for that application and paired prediction-level uncertainty. Marginal interval endpoints cannot be averaged to obtain such an interval.

E.2 Real-task optimization evidence

Table 11: Previously executed EEGNet training diagnostics, now summarized by task and branch. Counts refer to settings, identity blocks, and stimulus doses, not independent participants or studies. Objective reduction is the median initial-minus-final recorded training objective; these values are not test metrics and are not comparable performance measures across tasks.

Task	Branch	Fits	Epoch-one	Median epoch	Objective reduction
Emo	Direct	50	12	18	0.1104
Emo	Residual	50	10	17.5	0.1162
Urban	Direct	50	8	11	0.1922
Urban	Residual	50	7	7.5	0.2080

These aggregates come from the same verification record as the primary predictions, not from newly selected runs. The historical standard/compact EEGNet and matched shrinkage-calibration comparisons remain supporting evidence on their original supports. The repaired real SEED-V task provides a separate compatibility check, not a positive control for the current self-report regression task. None of these records spans full fine-tuning, all calibration algorithms, or repeated-training uncertainty. The scope of the matched increment remains the specified fitted predictor pair.

E.3 Post-primary partition, initialization, and fine-tuning checks

The alternative partition repeats assignment, fitting, aggregation, and verification with a new design seed; because the fitted data also change, it is a partition/refit sensitivity rather than an isolated partition effect. The three EEGNet schedules share the original assignments and four resource corners, changing only declared training initialization offsets. The LaBraM check uses the official checkpoint, freezes all but the final transformer block and normalization, fits a residual head, selects

Table 12: Post-primary robustness checks. Block-t intervals remain conditional on each fitted run. Alternative partitions and initialization schedules are sensitivity analyses, not independent cohorts.

Task	Check	Setting	Result
Emo	New blocks + refit	Six settings	max point shift 0.0135; 6/6 cover 0
Urban	New blocks + refit	Six settings	max point shift 0.0154; 6/6 cover 0
Emo	Three train seeds	EEGNet L1	[-0.0094, -0.0015]; 3/3 cover 0
Emo	Three train seeds	EEGNet L2	[-0.0018, +0.0003]; 3/3 cover 0
Urban	Three train seeds	EEGNet L1	[+0.0097, +0.0134]; 3/3 cover 0
Urban	Three train seeds	EEGNet L2	[+0.0013, +0.0109]; 3/3 cover 0
Emo	Last-block fine-tune	LaBraM residual	-0.0110 [-0.1041, +0.0820]
Urban	Last-block fine-tune	LaBraM residual	-0.0404 [-0.1159, +0.0351]

epochs on validation data, and then refits on training plus validation. Its 100-Hz analysis tensors are linearly resampled to the official 200-Hz patch interface; the Urban voltage conversion is checked from recorded metadata. All checks condition their block-t intervals on the resulting fitted predictions. They were specified after the primary results, cover neither a new cohort nor the full 5×5 fine-tuning surface, and cannot certify invariance to arbitrary partitions, optimizers, checkpoints, or adaptation methods.

E.4 Five-block interval validation

The production statistic first computes paired CCC differences within each identity block, then averages blocks and dose cells. It does not pool all trials before computing CCC. The new validation implementation preserves this order and uses the same global participant and stimulus multinomial weights across all blocks and cells. A draw invalid in any block is not repaired by omitting that block. Tests compare its cell estimates, uniform means, interval endpoints, and valid-draw counts with the production analyzer for both one- and two-target tasks.

The frozen validation configuration uses the exact complete diagonal test supports in the primary verification record: $6\times9, 6\times9, 6\times8, 6\times8, 6\times8$ for Emo, and $13\times12, 13\times11, 12\times11, 12\times11, 12\times11$ for Urban. It retains the full 5×5 dose grid, two targets for Emo and one for Urban. Four previously defined Gaussian scenarios exercise label-only cancellation, trial signal, participant-constant signal, and independent noise. Each scenario uses 300 fresh simulation replicates per task and 2,000 bootstrap draws. The primary block-t interval retains the crossed-bootstrap standard error and uses the five identity blocks as the independent aggregation units (df=4). Percentile, basic, and the earlier within-block t-normal intervals remain sensitivity checks. The correction was introduced after the initial interval diagnostics failed, but its critical value is fixed by the estimand topology and does not use the real EEG effect signs or magnitudes.

The estimand is the scenario's analytic population matched CCC difference, averaged uniformly over dose cells. Finite-block estimation bias is part of the stress test. Synthetic block resources are independent: matching this test topology does not reproduce all cross-block training-resource dependence, the actual EEG distribution, or retraining uncertainty. Coverage proportions require Monte Carlo uncertainty and do not establish validity outside the tested constructions. Executed results are recorded in the accompanying validation artifact.

Table 13: Coverage of the primary uniform-grid mean under the observed five-block topology (300 replicates per row). Brackets give Wilson Monte Carlo intervals for the primary block-t coverage.

Task	Scenario	Block-t [MC interval]	Percentile	Basic
Emo	Trial signal	0.993 [0.976, 0.998]	0.900	0.983
Emo	Participant constant	0.980 [0.957, 0.991]	0.903	0.910
Emo	Independent noise	1.000 [0.987, 1.000]	1.000	1.000
Emo	Labels only	1.000 [0.987, 1.000]	1.000	1.000
Urban	Trial signal	1.000 [0.987, 1.000]	0.927	0.977
Urban	Participant constant	0.983 [0.962, 0.993]	0.970	0.920
Urban	Independent noise	1.000 [0.987, 1.000]	1.000	1.000
Urban	Labels only	1.000 [0.987, 1.000]	1.000	1.000

All 2,400 simulation replicates completed without a failed run. The original 187,200 candidate rows and 62,400 derived block-t rows include every cell and uniform mean; they are not 249,600 independent simulations. Labels-only cancellation gives degenerate zero differences; its coverage of one is an algebraic check, not calibration of a nondegenerate interval. Independent noise produces no positive interval in these runs, which does not establish false-positive control for every null.

The earlier percentile failure is substantive. At Emo's zero-dose participant-constant cell, its population-target bias is -0.0756 and coverage is 0.193 (Monte Carlo interval $[0.153, 0.242]$); the earlier t-normal coverage is 0.887. The block-t construction raises the minimum cell coverage to 0.917, which still does not validate cellwise inference. Its uniform-mean coverage is 0.980–1.000 across the eight task–scenario combinations, with Wilson lower bounds of 0.957–0.987. These tests support the primary uniform-grid mean under the simulated topology, not every cell, the actual EEG distribution, arbitrary signal families, or retraining uncertainty. Resource identity and descriptive paired differences do not depend on a nominal significance threshold; no equivalence claim is made.

Historical Supporting Analyses

These analyses retain their original test supports and estimators. They are supporting records, not independent replications of the current primary grid. The current design and diagnostics precede this historical supplement.

F Data Sources and Trial Reconstruction

F.1 Version-pinned sources

OpenAffect-EEG uses fixed OpenNeuro snapshots rather than a floating latest version: MusicEEG ds002721 version 1.0.3, DENS ds003751 version 1.0.6, EmoEEG-MC ds005540 version 1.0.7, and external validation ds006866 version 1.0.0 [11, 26, 29, 38]. Snapshot manifests, byte counts or annex keys, and derived trial-table hashes record the source state. Across the 4.0 datasets we reconstruct 39642.0 trials from 279.0 dataset-qualified participants; 38369.0 trials have complete harmonizable continuous targets. The core tier contains 131.0 participants, 4135.0 trials, 369.0 qualified stimulus identities, and 2865.0 supervised trials.

The DENS gap originates from unreadable public signal windows rather than an explicit target filter, but the resulting complete-case set is not ignorable. Only 162.0 of 365.0 label-eligible trials are available (missing fraction 0.5562); 7.0 participants have no available trial, and the available-versus-missing valence standardized mean difference is 0.1895. Missingness summaries and probe seen-class fractions are published per dataset and seed. Subject probes in the LaBraM case study have complete test coverage; permutation labels are shuffled only in the training partition while the assignment, features, estimator, and test labels remain fixed.

MusicEEG. Participants listened to film-music excerpts and completed eight self-report items [10]. The benchmark retains 1,240 task trials. Public events do not directly provide canonical audio filenames, so `stimulus_uid` is aligned from event codes to the public OSF Set 1 after checking uniqueness, code range, and the availability of audio and normative ratings for every candidate. Because the README, event description, and early paper do not fully agree on the material set, this mapping is labeled `audited_inference`, not `source_verified`. The OpenNeuro snapshot declares CC0 and the companion OSF resource declares CC BY 4.0; audio is not redistributed.

DENS. DENS uses naturalistic videos and collects dimensional and categorical ratings [2]. Events and behavior are joined by normalized stimulus name, never row position. Several public `.fdt` lengths disagree with their `.set` headers; unreadable full windows are recorded through an availability mask rather than padded, wrapped, or treated as valid EEG. Public events contain twenty local stimulus names, but the snapshot lacks the documented stimulus and code directories, and the cited AFDI entry cannot be reliably linked to those local identities. They are therefore valid grouping keys but not claims of recovered source films. Conflicting CC0 and ODbL/DbCL statements are handled under the more conservative attribution and share-alike boundary.

EmoEEG-MC. The dataset combines video viewing and text-guided imagery [39]. A published behavioral `video_name` field repeats one erroneous value for some participants and cannot be joined by row order. The supervised subset is restricted to globally verifiable stimulus codes, and context direction is cross-checked against source EDF trigger types. Other reconstructed trials remain in provenance with unavailable targets. Public DE arrays, reordered EEG, model features, and checkpoints are excluded from the anonymous package.

External emotion regulation. ds006866 studies social and nonsocial images under neutral viewing, negative viewing, and reappraisal conditions [27]. It contributes 148.0 participants and 35507.0 image-presentation trials, of which 35504.0 have complete valence and arousal. Public events recover condition but not trial-level image identity, and EEGLAB annotations do not restore it. Stimulus and joint holdout are consequently blocked by source metadata. The OpenNeuro snapshot declares CC0; companion metadata declare CC BY 4.0, while source images can have separate terms. Neither images nor a substitute media index are distributed.

Table 14: Source and redistribution boundaries. License entries describe the fixed source metadata; companion media and code retain separate terms.

Dataset	OpenNeuro snapshot	Companion boundary	Benchmark release
MusicEEG	CC0	OSF metadata/audio: CC BY 4.0	Metadata, splits, scores
DENS	CC0 field; ODbL/DbCL conflict	Videos not mirrored	Conservative attribution
EmoEEG-MC	CC0	Stimuli/code/checkpoints separate	No EEG, DE, media, weights
ds006866	CC0	OSF table: CC BY 4.0; source images separate	No EEG or images

Table 15: Observed-data and probe-coverage disclosure. EEG feature counts are applied after a frozen trial assignment; missing signals never redefine a split. Probe coverage is the test fraction whose class was observed in fit.

Dataset	Labeled	EEG features	Main role	Stimulus probe
MusicEEG	1,240	1,240	Transparent spectral audit	0.8708
DENS	365	162	Transparent, missingness-limited	0.9826
EmoEEG-MC	1,260	1,260	DE and foundation case study	1.0000
ds006866	35,504	35,504	Condition-level external audit	N/A

G Target Harmonization

MusicEEG pleasantness and energetic-arousal reports define valence and arousal; the other datasets use their released valence and arousal fields. MusicEEG, DENS, and ds006866 map an original one-to-nine scale by $(x-1)/8$; EmoEEG-MC maps zero-to-seven by $x/7$. Original values, scale labels, and transformation expressions remain in trial metadata or result provenance.

The supervised target is the participant's trial-level report. Nominal stimulus category, normative rating, and experimental condition can be inputs, stratification variables, or audit labels, but do not replace the participant target. Trials missing either continuous dimension remain in metadata and are excluded only from metrics requiring both dimensions.

H Split Protocols

Single-factor protocols and the optimistic control. Complete trials are assigned before windowing. Random-trial, subject, and stimulus holdout use 70/10/20 train/validation/test proportions within each dataset, with at least one identity reserved for validation and test in small groups. Random-trial splitting isolates only `trial_uid` and records subject and stimulus overlaps. Strict single-factor splits require an empty train–test intersection for the held identity.

Joint holdout. Subjects and stimuli are independently assigned 60/20/20. A trial is retained only when both identities receive the same partition; cross-partition edges are marked `excluded`. This sacrifices coverage in sparse subject–stimulus graphs to obtain a genuine double cold start. The ratio was fixed before model evaluation because smaller validation identity sets did not provide adequate validation coverage across seeds.

Support-matched identity-overlap control. For each joint-holdout seed, the control freezes the exact test trial IDs and matches the joint split's train and validation row counts. It then samples those training-side rows so that both test subjects and test stimuli occur in the fit partition. Consequently, the control preserves test support and sample counts while changing training-side identity overlap and the associated graph. It is an estimand for the practical effect of available identity structure, not a claim that all higher-order graph properties are identical. Across seeds, test sets contain 32.0–56.0 trials from 4.0–7.0 participants and exactly 8.0 held-out stimuli.

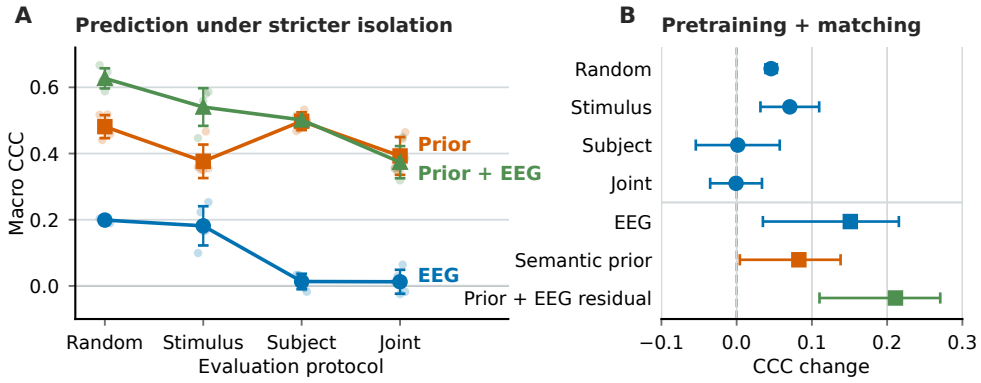


Figure S2: Protocol-level audit retained as supporting evidence. **A:** five fixed split seeds and their mean macro-CCC for semantic prior, direct EEG, and prior-plus-EEG residual; random-trial splitting is an optimistic control. **B:** pretrained versus fixed-random LaBraM comparisons and support-matched identity-overlap effects. These estimates motivate the response-surface analysis but do not by themselves separate participant calibration from repeated-stimulus exposure.

Context and dataset holdout. Directional EmoEEG-MC transfer first holds out subjects. Only source-context trials enter training and validation, and only target-context trials from unseen subjects enter test. Video-to-imagery and imagery-to-video are evaluated separately. Context and dataset holdout treat a complete domain as a group.

All main representation audits use 5.0 fixed seeds. Each assignment stores trial roles and an audit JSON. Builders check UID uniqueness, nonempty identity fields, legal roles, and the required empty identity intersections. Models apply target and feature-availability masks only after loading an assignment, so representation missingness cannot silently alter the split.

H.1 Held-out deployment-selection utility

To test a practical consequence rather than assume that ranking changes help, we compare a validation-cell selector with one global validation-mean selector. Both receive identical predictions from seven candidates: a personalized EEG-free prior and direct/combined predictions from spectral features and frozen pretrained/random LaBraM. Candidates, dose grids, role seed and rules are fixed before this analysis. No candidate is removed based on test performance.

Participant and stimulus identities have disjoint fitting, inner tuning, selector-validation and test roles. All trials containing the other phase's participant or stimulus identities are quarantined, including resource trials. Assignments use constant placeholders for target-block labels. Ridge tuning uses only inner tuning outcomes; selector choices are hashed before test evaluation. Five resource seeds share the same role partition and target support, not five independent cohorts. The EEG-free and EEG predictors receive identical calibration labels and stimulus-label resources.

EmoEEG-MC has 48 trials from six participants and eight stimuli in each of the two phases. MusicEEG has 25 validation trials (six participants, eight stimuli) and 23 test trials (six participants, nine stimuli), after maximum-dose feasibility filtering. The sparse support limits precision and coverage. All 2,000 paired crossed draws are valid. They condition on fits and validation choices, not selector-training uncertainty. Source, assignment and prediction hashes are verified, and a separate centered-moment implementation reproduces the scores and intervals. This is developer verification, not an independent human reproduction.

Neither primary CCC interval excludes zero. Both secondary MAE differences are small and negative, but no practical-importance threshold was established; they cannot replace the primary endpoint or identify EEG added value. The global choices are random-LaBraM combined prediction for EmoEEG-MC and spectral combined prediction for MusicEEG. These choices describe this candidate set and validation block, not population-level model superiority. This analysis uses datasets

Table 16: Held-out deployment-selection comparison. Both policies use the same validation predictions. Differences are conditioned minus global, uniformly averaged across resource cells; intervals are paired crossed 95% intervals conditional on fits and choices. CCC is primary; MAE is secondary and lower is better.

Dataset	Metric	Global	Conditioned	Difference [interval]
EmoEEG-MC	CCC	0.34221	0.34477	+0.00256 [-0.00113, 0.00601]
EmoEEG-MC	MAE	0.19523	0.19464	-0.00059 [-0.00103, -0.00013]
MusicEEG	CCC	0.24712	0.24600	-0.00113 [-0.01101, 0.01027]
MusicEEG	MAE	0.29405	0.29021	-0.00384 [-0.00675, -0.00094]

already examined during project development: role isolation protects the current selection procedure, but is not untouched external confirmation.

H.2 Identity-exposure response surface

Identity availability is not represented as one binary variable. A deployment state can separately specify whether the participant ID is known, whether unlabelled target-participant data are available, how many labelled calibration trials are available, whether calibration comes from the same session, and whether the test stimulus, dataset, and device occurred during fitting. The present response surface varies exactly two labelled resources: p, complete calibration trials per target participant, and s, other-participant ratings per test stimulus (or experimental condition where stimulus identity is unavailable). Released normative stimulus metadata, where present, is held fixed and reported as a separate prior rather than counted as s. It fixes unlabelled target-participant adaptation at zero. Session and device are not separately isolated by the available metadata and are therefore blocked claims, not implicit successes.

For each seed, we begin from one strict participant–stimulus joint assignment. The test set is restricted once to participants and stimuli supporting the predeclared maximum doses; its trial IDs are then hash-fixed for every (p, s) cell. Participant calibration candidates contain a test participant under a non-test stimulus. They never enter population-model fitting. Repeated-stimulus candidates contain a test stimulus rated by a non-test participant. Dose samples are deterministic and nested, so every lower dose is a subset of every higher dose on the same axis.

Adding s repeated-stimulus rows would otherwise increase sample size. Each added row therefore replaces one base-training donor selected deterministically within context and nearest in the two training targets. Population-training size is exactly constant across cells, while validation and test trials are unchanged. Donor selection reads training labels only. The intervention intentionally changes which stimulus identities are represented during fitting; it does not claim all graph or target-distribution properties are causally matched.

Let f_s be a population predictor fit with stimulus exposure dose s. For a target participant i, labelled calibration produces only the post-fit offset

$$\widehat{\boldsymbol{\delta}}_{i,p} = \frac{1}{p} \sum_{j \in \mathcal{C}_{i,p}} \left(\mathbf{y}_{ij} - f_s(\mathbf{x}_{ij}) \right), \tag{5}$$

with zero offset at $p = 0$. Test prediction is $f_s(\mathbf{x}_{i,t}) + \widehat{\boldsymbol{\delta}}_{i,p}$. Thus a nonzero p cell is explicitly a known-participant, labelled-calibration result; it is not subject independent. The same construction is applied to the prior-plus-EEG predictor and to the EEG-free prior with exactly the same calibration rows. Population and personalized outputs are stored separately.

Matched model extension and reproducibility. The EmoEEG-MC extension evaluates EEGNet and frozen EMOD on every existing assignment, asserting identical test IDs, targets, priors, and calibrated priors. EEGNet selects its epoch count by validation MAE, then refits on train plus validation; target-participant calibration does not update network weights. Direct and residual models are fitted separately. Population fits are reused across participant doses only after verifying unchanged fit assignments. The compact model uses the previously specified custom EEGNet-style architecture with zero adversarial loss. All preprocessing statistics are estimated from its fit data.

Table 17: Maximum-resource predictions with resource-matched controls. P: prior; PC: calibrated prior without EEG; E: prior plus population EEG residual; EC: calibrated E. Brackets are paired crossed-identity 95% pointwise intervals. These are not equivalence tests.

Representation	P	PC	E	EC	EC − PC [95% CI]
EmoEEG-MC (p=8, s=8)					
DE	0.4809	0.5863	0.4416	0.5599	-0.0265 [-0.0640, 0.0178]
EEGNet	0.4809	0.5863	0.4863	0.5955	0.0091 [-0.0059, 0.0307]
LaBraM pretrained	0.4809	0.5863	0.4676	0.5748	-0.0115 [-0.0496, 0.0298]
LaBraM random	0.4809	0.5863	0.4629	0.5711	-0.0152 [-0.0553, 0.0179]
CBraMod pretrained	0.4809	0.5863	0.4590	0.5686	-0.0177 [-0.0472, 0.0235]
CBraMod random	0.4809	0.5863	0.4817	0.5840	-0.0023 [-0.0327, 0.0262]
EMOD frozen	0.4809	0.5863	0.4738	0.5819	-0.0045 [-0.0359, 0.0258]
EMOD random	0.4809	0.5863	0.4530	0.5687	-0.0176 [-0.0669, 0.0278]
MusicEEG (p=8, s=4)					
Band power	0.1740	0.2930	0.1799	0.2988	0.0058 [-0.0131, 0.0253]
LaBraM pretrained	0.1740	0.2930	0.2085	0.3070	0.0141 [-0.0332, 0.0569]
LaBraM random	0.1740	0.2930	0.1805	0.2941	0.0011 [-0.0531, 0.0587]

EMOD uses official repository revision `3a219f51ba4f90b9e2d22824cfb90dde77a9875c`. Its checkpoint has legacy pooling/projection names; the adapter records a one-to-one name mapping and then requires a strict match of all tensors. It uses the benchmark's 100-Hz raw archive, three nonoverlapping ten-second crops, and mean pooled official projection embeddings. Channels absent from the official channel vocabulary are excluded and listed in the extraction manifest. Input scaling/preprocessing and head adaptation differ from the paper's full fine-tuning pipeline. This audits the supplied frozen weights, not a claim that EMOD's original reported SOTA performance was reproduced. Neither third-party weights nor raw feature arrays are redistributed.

Failed native-task learning check. A separate bounded SEED-V check used the official SEEDV model branch and checkpoint backbone with 62 channels, 200 Hz, microvolt inputs and one-second windows. Within each participant/session, the first five, middle five and last five trials formed train, validation and test. Across 48 source files there were 117,744 windows; 40,352 test windows represented 240 trials from 16 participants. A single fixed seed, 100 epochs, and validation-kappa checkpoint selection yielded test balanced accuracy 0.20 and kappa 0: all test windows received one class. Prepared-input hashes, test identities, finite gradients and recomputed metrics were verified, but native-task learning was not recovered. The released SEEDV branch uses a large multilayer head, whereas the paper describes a linear classifier; the per-task selected fine-tuning recipe is not fully specified. This failed compatibility check neither reproduces nor refutes the published scores. It cannot validate the frozen regression audit as a full SOTA test. CUDA pooling backward is nondeterministic despite fixed seeds; this is recorded with the aggregate training history.

Training-only diagnosis and repair. We retained that failed run and reproduced its numerical failure on 50 fixed training windows, ten per class, without validation/test-based parameter selection. With classifier learning rate 10^{-3}, the first update increased the first dense layer's output standard deviation from 0.773 to 32.280; by step 20, the second hidden ELU produced constant outputs. The official SEEDV head flattens 39,680 features into a 2,480-unit layer. Keeping the architecture, inputs, initialization, optimizer and backbone learning rate 5×10^{-4} unchanged, reducing only the classifier learning rate to 10^{-5} prevented saturation and fitted the diagnostic batch by step 50. This establishes a configuration-specific numerical remedy, not an emotion-mechanism finding. All 48 training-label files matched the official constants. Reprocessing 100 training windows across five source files, including the format-fallback and largest-amplitude cases, matched the prepared float32 inputs exactly; this is selected-file parity, not exhaustive independent data validation.

The repaired run completed the fixed 100-epoch budget. Validation kappa selected epoch 21, after which the test set was evaluated once. Recomputed test BACC was 38.62%, kappa 0.2348 and weighted F1 39.43%; all five classes were predicted. The published five-run BACC is $41.24 \pm 0.86\%$ [9], leaving a 2.62 percentage-point gap to that mean. This single-seed check demonstrates recovered learning, not statistical equivalence or a five-run reproduction. It uses within-participant classification, not the self-report exposure response surface, and cannot establish unseen-user transfer or pretrained

benefit without a matched random-initialization comparison. Aggregate traces, input/source hashes and a fail-fast input-sensitivity check accompany the repair. The original failed results remain archived.

Standard implementation and stronger calibration control. The added standard regression model uses Braindecode 1.2.0 EEGNet with $F_1 = 8$, $D = 2$, $F_2 = 16$, temporal kernels 64/16, dropout 0.25, and two raw regression outputs. AdamW uses learning rate 0.001 and weight decay 0.0001; validation MAE selects among at most 100 epochs with patience 12. All 25 population states and 125 dose cells use the archived assignments. Its 50 direct/residual fits have finite logged objectives and gradients; selected epochs range from 1 to 24, with 14 fits selecting epoch one. Synthetic learning and training-only tiny-subset fitting checks pass, but do not demonstrate optimal benchmark tuning or held-out emotion information.

An additional empirical calibration control estimates within-participant residual variance σ^2 and nonnegative noise-corrected between-participant mean variance τ^2 from population-fit data. It multiplies every calibrated offset by $w_p = p\tau^2/(p\tau^2 + \sigma^2)$, with $w_0 = 0$. These are heuristic prediction weights, not identified latent variance components. The same weights apply to PC and EC. Table 18 reports uniform cell means; the historical dose-weighted integrals remain separate secondary summaries. Neither calibration rule establishes a positive surface-average EEG increment for the displayed models at the reported precision.

Revised uncertainty. The paired tables use 2,000 shared crossed-identity draws across fixed splits. The same draws compare every predictor, dose, and model; a degenerate draw is not repaired by silently dropping a seed. The tables expose valid draw counts. The legacy resource-effect table below preserves the original seedwise bootstrap for traceability and is not the revised estimator used in the main text. In particular, MusicEEG's revised joint interval crosses zero. For ds006866, each fixed split holds out one condition; its condition factor cannot support a within-split resampling claim about new conditions.

MusicEEG's frozen grid is five participant doses by four stimulus doses, ending at $s = 4$. Raising s to eight cannot preserve the same test support in four of the five seeds: their available other-participant observations are insufficient. Only one seed supports that extension without dropping test trials. We do not fill unavailable cells or compare a changed test set as if it were the same response surface.

The artifact emits one deterministic claim card per cell. It records ID and label requirements and blocks latent-emotion disentanglement, causal identity removal, unseen-participant claims when $p > 0$, unseen-stimulus claims when $s > 0$, and session/dataset/device transfer unless those factors are separately held out. This is rule-based compilation from split lineage, not free-form language-model judgment.

I EEG Representations and Models

Transparent spectra. Core transparent features use delta, theta, alpha, beta, and gamma DE or Welch log power. Global views aggregate channels; channel views retain the channel-by-band layout. Standardization is fitted on training data only. Ridge regularization is selected by validation MAE and refitted on train plus validation. Single-band, leave-one-band, single-region, and leave-one-region views test predictive sensitivity, not source localization or causal anatomy.

CBraMod. Official pretrained and same-architecture random encoders receive identical EmoEEG-MC trials. Each 30-second, 64-channel, 200-Hz trial is organized into one-second patches; channel and time tokens are mean-pooled into a 200-dimensional trial representation. Source commit, checkpoint SHA-256, trial UIDs, and feature hashes accompany the artifact. The encoder is frozen and only the common downstream head is trained.

LaBraM. LaBraM uses the official channel-location map and base checkpoint. Thirty one-second patches are processed as three eight-patch windows and one six-patch window, then pooled with patch-count weighting into a 200-dimensional trial representation. Pretraining masks and decoder heads are excluded, while the official final normalization path is retained. The random control instantiates the same architecture without loading a checkpoint.

Table 18: Implementation and calibration sensitivity. Entries are uniform-grid mean matched EEG added value, $\mathrm{CCC}(EC) - \mathrm{CCC}(PC)$, with 95% paired crossed intervals conditional on fitted predictions. Shrinkage weights are estimated only from population-fit residuals and shared by EEG and non-EEG offsets. These are not maximum-dose scores or equivalence tests.

Dataset	Representation	Mean offset	Shrunk offset
EmoEEG-MC	Compact EEGNet-style	0.0089 [-0.0034, 0.0254]	0.0067 [-0.0064, 0.0253]
EmoEEG-MC	Standard EEGNet	0.0006 [-0.0036, 0.0054]	0.0006 [-0.0044, 0.0060]
EmoEEG-MC	LaBraM	-0.0100 [-0.0392, 0.0279]	-0.0098 [-0.0419, 0.0308]
EmoEEG-MC	Frozen EMOD	-0.0001 [-0.0280, 0.0247]	-0.0005 [-0.0302, 0.0263]
MusicEEG	Band power	0.0042 [-0.0094, 0.0188]	0.0052 [-0.0269, 0.0335]
MusicEEG	LaBraM	0.0182 [-0.0161, 0.0452]	0.0247 [-0.0176, 0.0568]

Table 19: Companion error contrasts for the same matched resources. Entries are uniform-grid mean $\mathrm{MAE}(EC) - \mathrm{MAE}(PC)$ with paired crossed 95% intervals. Negative values favor the EEG branch; positive values mean larger error. These marginal intervals do not control error across all representations and calibration variants.

Dataset	Representation	Mean offset	Shrunk offset
EmoEEG-MC	Compact EEGNet-style	-0.0008 [-0.0030, 0.0011]	-0.0002 [-0.0032, 0.0023]
EmoEEG-MC	Standard EEGNet	0.0004 [-0.0004, 0.0012]	0.0005 [-0.0005, 0.0015]
EmoEEG-MC	LaBraM	0.0059 [0.0001, 0.0114]	0.0056 [-0.0009, 0.0122]
EmoEEG-MC	Frozen EMOD	0.0032 [-0.0025, 0.0100]	0.0029 [-0.0029, 0.0099]
MusicEEG	Band power	0.0013 [-0.0021, 0.0044]	0.0033 [-0.0045, 0.0108]
MusicEEG	LaBraM	-0.0006 [-0.0071, 0.0067]	-0.0010 [-0.0100, 0.0096]

Historical compact EEGNet-style control. The custom task-specific control is trained from EmoEEG-MC waveforms resampled with anti-aliasing to 100 Hz. Each epoch uses one deterministic 10-second training crop; validation and test average three nonoverlapping crops. The network uses temporal convolution, depthwise spatial convolution, separable temporal convolution, and an eight-dimensional latent. It tests architecture dependence, not an exhaustive architecture search.

J Predictors, Debiasing, and Identity Probes

Predictors. `direct_eeg` predicts normalized targets from EEG. `semantic_stimulus_prior` combines one-hot nominal category, context, and source stimulus-intensity code with lowercased TF–IDF description unigrams/bigrams (at most 1,024 features, sublinear term frequency). A Ridge model is fitted to one mean target row per training stimulus; alpha is selected from $\{0.01, 0.1, 1, 10, 100, 1000\}$ by trial-level validation MAE. The intensity code identifies source stimulus variants and never uses a held-out participant's rating. The historical registry identifier `disentangled_prior_plus_eeg_residual` uses a training-only stimulus report mean for seen stimuli and the semantic estimate for unseen stimuli, forms leave-one-trial-out residual targets, predicts them from EEG, and adds the hybrid prior. All branches share the assignment and validation rule; none reads test targets. The identifier denotes arithmetic residualization and does not claim causal or latent-state disentanglement.

Debiasing counterfactuals. We evaluate stimulus centering, conditional covariance penalties, response-preserving stimulus-subspace projection, gradient-reversal residual networks, and adversarial EEGNet. Every method contains a zero-strength setting; adversarial strength, projection rank, and epoch are selected by validation residual MAE only. Test-time residual prediction does not use stimulus ID. A successful counterfactual must jointly improve target performance and reduce the relevant identity effect. Variance shrinkage or inconsistent seed direction is reported as a negative result.

Identity probes. Dataset identity is predicted under subject holdout; stimulus identity under subject holdout is evaluated only on classes observed in training; subject identity under stimulus holdout is

Table 20: Legacy fixed-support identity-exposure summary (original seedwise bootstrap, retained for traceability). Values are macro-CCC changes from the zero-resource cell; brackets are crossed participant–stimulus 95% intervals. For ds006866, S denotes experimental-condition exposure because source metadata do not expose trial-level image identity.

Dataset	Representation	Zero	Maximum	Surface AUC	ΔP	ΔS	$\Delta(P{+}S)$
EmoEEG-MC	LaBraM pretrained	-0.0052	0.5748	0.4604	0.1788 [0.0712, 0.2307]	0.4728 [0.2900, 0.5420]	0.5800 [0.4072, 0.6219]
MusicEEG	LaBraM pretrained	0.1026	0.3070	0.2302	0.0953 [-0.0097, 0.2037]	0.1060 [-0.0950, 0.2798]	0.2045 [0.0157, 0.3320]
ds006866	Band power	–	–	–	0.2856 [0.2132, 0.3461]	-0.0014 [-0.0071, 0.0042]	0.3256 [0.2694, 0.3685]

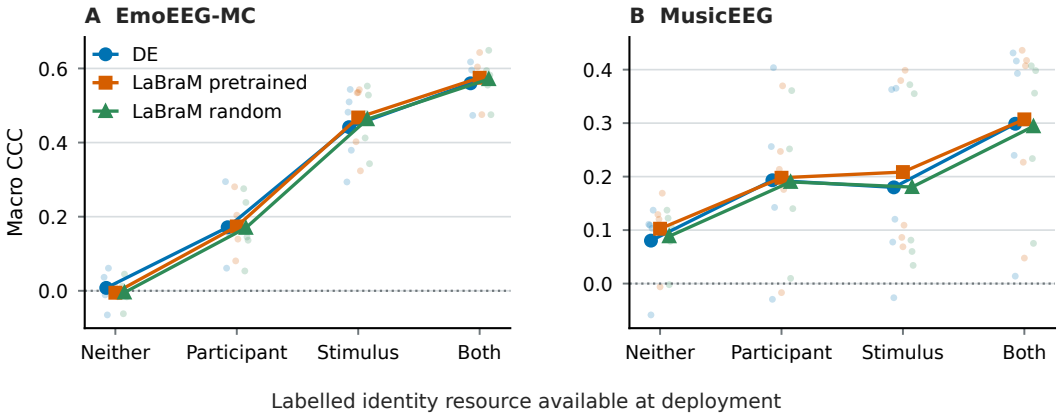


Figure S3: Deployment corners for the three main representations. Large markers and connecting lines show seed means; faint points are the five fixed split seeds. "Participant" uses the maximum labelled calibration dose with unseen stimuli, "Stimulus" uses the maximum repeated-stimulus dose with unseen participants, and "Both" uses both resources. The large MusicEEG spread is why endpoint means are not treated as independent replications.

likewise evaluated on training-seen people. Each probe uses a matched training-label permutation and reports balanced accuracy, class count, uniform chance, coverage, and per-seed values.

Subject-subspace intervention. For each stimulus-holdout seed, between-subject centroid differences are estimated on training data, decomposed by SVD, and the leading directions are removed from fit and test LaBraM representations. The requested removal grid is $\{0, 1, 2, 4, 8, 16, 24, 29\}$. At every rank, we also remove the same number of training-estimated PCA directions and seeded random orthonormal directions. Semantic, direct, residual, and subject-probe heads are refitted with the same validation-only selection. The zero-removal point is the nested control. This intervention destroys linearly decodable subject structure but can also remove participant-related affect variation; it is not interpreted as a pure nuisance projector.

K Statistical Analysis

Continuous predictions report valence, arousal, macro CCC, MAE, and RMSE. Identity tasks report balanced accuracy and, where available, macro-F1. Model comparisons are paired on identical trial-level predictions. In the legacy non-surface analyses, the hierarchical bootstrap first resamples evaluation seeds and then complete held-out identity clusters; joint holdout separately resamples crossed subject and stimulus factors. Those historical intervals use 2,000 resamples and a fixed statistical seed. This is not the revised primary response-surface estimator, which holds split seeds fixed and shares crossed identity weights across every model/cell.

With 5.0 seeds, the exact two-sided sign-flip test has a minimum nonzero p-value of 0.0625. We therefore interpret bootstrap intervals as effect uncertainty and the sign-flip test as a finite-seed resolution statement. Historical targeted robustness analyses comprise the support-matched CCC contrast, subject-subspace endpoint CCC contrast, unseen-stimulus LaBraM pretraining contrast, and external condition-prior paired contrasts. These were selected during benchmark development and reviewer-driven audit rather than preregistered. Other reported cells are exploratory. Seeds

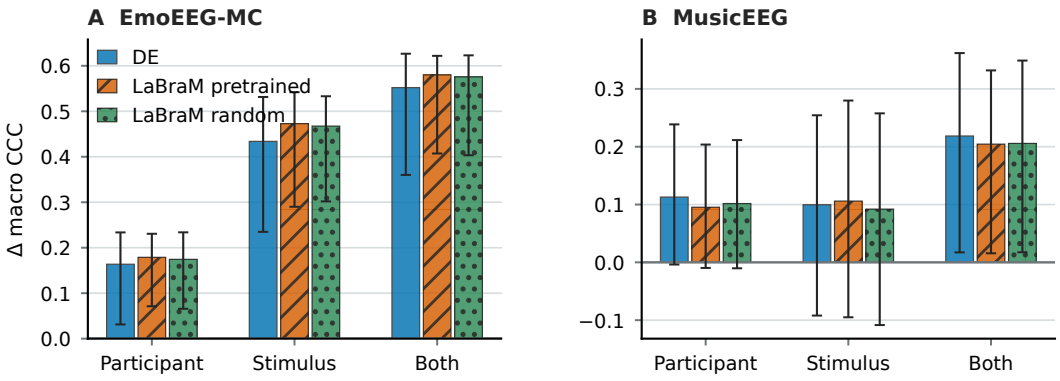


Figure S4: Maximum-dose minus zero-resource changes. Bars distinguish transparent features, pretrained LaBraM, and fixed-random LaBraM; whiskers are crossed participant–stimulus bootstrap 95% intervals. EmoEEG-MC supports positive marginal and joint changes under the legacy estimator. The revised shared-identity interval for MusicEEG's joint change crosses zero; this legacy plot is not evidence of a confirmed external resource effect.

Table 21: Dataset–protocol applicability matrix. "Blocked" denotes missing source identity metadata, not model failure.

Dataset	Random	Subject	Stimulus	Joint	Context	Dataset
MusicEEG	Yes	Yes	Yes	Yes	N/A	Yes
DENS	Yes	Yes	Yes	Yes	N/A	Yes
EmoEEG-MC	Yes	Yes	Yes	Yes	Yes	Yes
ds006866	Yes	Yes	Blocked	Blocked	Yes	N/A

represent Monte Carlo split variability; participant, stimulus, or crossed cluster resampling supplies the corresponding generalization uncertainty. Intervals are pointwise and are not presented as simultaneous family-wise coverage.

Under the same support-matched contrast, the semantic stimulus prior changes by 0.0827 CCC and the hybrid stimulus prior by 0.0991. These branches are reported separately so the full-pipeline effect is not attributed to EEG alone. Component ablations further separate nominal category, context, source intensity code, and description text (Table 26).

The strong nominal-category effect under unseen-stimulus and joint holdout shows that the semantic prior's transferable score is largely experiment-label structure. Description text contributes under subject holdout, whereas context, intensity, and description do not have stable independent joint-holdout effects. These are predictive ablations, not causal interventions on stimulus content.

L External Validation Details

The initial subject-holdout analysis found that the condition prior exceeded EEG-only macro-CCC by 0.3457, with interval [0.304, 0.3884]. The v2 audit extends this case to exactly matched bandpower, pretrained-LaBraM, and fixed-random-LaBraM representations under five protocols. Table 28 reports every frozen cell rather than only the favorable repeated-participant setting.

For repeated participants, the hybrid improves over the prior in both directions and all five participant-clustered MAE intervals lie below zero for each representation. Under joint participant–context holdout, no LaBraM hybrid interval establishes an improvement over the prior. Pretrained-minus-random hybrid CCC is -0.0008 and -0.0082 across the two joint directions, with only two of five seeds positive in each. Thus the strict result does not support a pretraining-specific residual advantage.

The near-perfect repeated-participant probes establish information available to a fixed linear readout; they do not show that the affect head causally uses identity. Public events expose condition but not

Table 22: Executed analysis coverage. This table is intentionally distinct from protocol applicability: "supported" does not imply that every model was run in that cell. V/I denotes directional video–imagery transfer.

Dataset	Representation or analysis	Executed protocols
EmoEEG-MC	DE, CBraMod, LaBraM affect heads	Random, subject, stimulus, joint, V/I; exposure grid
EmoEEG-MC	EEGNet	Joint; complete exposure grid
EmoEEG-MC	EMOD frozen and random	Complete exposure grid
MusicEEG	Band power, LaBraM pretrained/random	Complete feasible exposure grid
MusicEEG, DENS, EmoEEG-MC	Transparent identity probes	Dataset, subject, stimulus as applicable
ds006866	Channel/global bandpower	Subject holdout and condition probe

Table 23: Representation controls and training state.

Representation	Trial view	Pretraining control	Downstream state
DE / Welch	Global, channel, band, region	N/A	Validation-selected Ridge
CBraMod	Frozen 200-D embedding	Same architecture, random	Frozen encoder
LaBraM	Frozen 200-D embedding	Same architecture, random	Frozen encoder
EEGNet	Raw deterministic crops	Zero adversary	End-to-end control
EMOD	Frozen 128-D embedding	Same architecture, random	Validation-selected Ridge

trial-level image identity, so stimulus holdout remains blocked. Header-only conformance confirms the same ordered 64-channel EEG set in all 148 recordings; 146 match the base source pattern and two contain one additional, predeclared EMG channel.

M Authorized SEED-V Protocol-Audit Case Study

SEED-V is a stimulus-induced five-class EEG and eye-movement study with 16 participants, three sessions, and fifteen film clips per session [20]. We report a separately authorized raw-EEG analysis as an independent case study rather than a fifth OpenAffect-EEG source. Its source recordings, feature arrays, media, trial predictions, participant codes, and project paths are not redistributed. The public artifact contains only the source-safe aggregate in `paper/generated/seedv_case_study.json`, its input and code hashes, and this interpretation boundary.

The authorized project compared a raw-temporal baseline against a pre-existing SDSA variant under a strict leave-one-participant-out protocol and a joint participant–stimulus protocol. Table 30 reports all frozen cells rather than selecting the favorable joint cell. The strict SDSA contrast is negative, and both bootstrap intervals include zero. Its frozen decision gate therefore fails; the contrast is evidence of protocol sensitivity, not a model-improvement claim. Separately, the authorized project's strict-to-PaperStyle ladder rises from 0.2786 to 0.5922, so the large protocol gap cannot be attributed to a model variant alone [20].

N Independent MusicEEG Protocol Audits

To test whether the EmoEEG-MC representation finding is confined to one visual paradigm, we run two prespecified audits on independent MusicEEG [10]. Each uses 1,240 labelled 12-second music trials from 31 participants, five frozen complete-trial splits, and source-safe aggregate reporting. First, a transparent channel log-bandpower Ridge readout reaches macro CCC 0.1031 under random-trial evaluation, 0.0175 with unseen participants, 0.1130 with unseen music but known participants, and 0.0227 under joint participant–music holdout. The paired random-minus-joint difference is 0.0804 (seed SD 0.0357). Thus holding out music alone does not substitute for holding out participants in this auditory source.

Table 24: Optimization and selection budget. Every choice is made on the validation partition and then reused for the paired test comparison.

Component	Validation search	Training budget
Ridge prior/EEG/residual heads	Six alphas from 10^{-2} to 10^{3}	Closed-form per seed/protocol
Linear nuisance controls	Five strengths from 0 to 1	Closed-form per strength
Gradient-reversal residual	Six adversary strengths	At most 200 epochs; patience 25
Adversarial EEGNet	Three adversary strengths	At most 100 epochs; patience 12
Subject-subspace audit	Eight ranks, three direction types	Head refit at every point

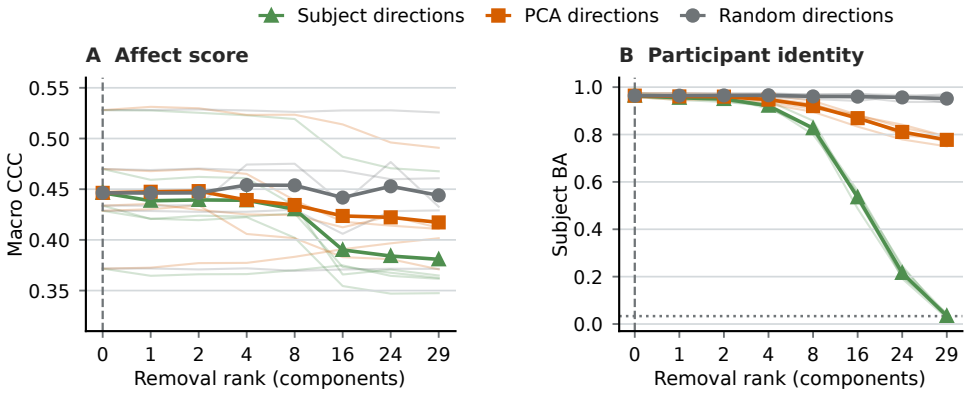


Figure S5: Subject-subspace intervention audit. **A** plots affect macro-CCC and **B** participant balanced accuracy across the fixed removal-rank grid. Faint trajectories are the five evaluation seeds; opaque lines and markers are their means. Triangles remove subject-aligned directions, squares remove equal-rank PCA directions, and circles remove equal-rank random directions. The vertical dashed line is the nested zero-rank control; the horizontal dotted line in **B** is chance. The figure exposes an accuracy–identity trade-off and does not identify participant structure as a pure nuisance factor.

The same bandpower representation identifies training-seen participants under unseen music at balanced accuracy 0.9995 (permuted 0.0431; 31-class uniform chance 0.0323). In contrast, the seen-class music probe under unseen participants is 0.0000 balanced accuracy (permuted 0.0033; 78.8–91.3% test coverage across seeds). These probes establish available information under their stated coverage, not causal use by the affect predictor.

The self-report diagnostic also constrains interpretation. Leave-one-rating-out same-music consensus CCC is 0.1437 for valence and 0.1634 for arousal (1,123 supported ratings each). Descriptive in-sample additive models assign more R-squared to music identity (0.3579 and 0.3484) than participant identity (0.1170 and 0.1782); they are neither test–retest reliability estimates nor causal variance decompositions.

Second, we apply the frozen official LaBraM base encoder to exactly the same MusicEEG split family. Each trial uses its 19 recorded 10–20 channels and its observed event window; MNE volts are converted to microvolts, common-average referenced within trial, resampled to 200 Hz, and final tokens pooled. There is no padding, cross-trial window, fine-tuning, or feature release. Pretrained and fixed-random initializations use the same 200-dimensional Ridge readout, manifests, five split seeds, and nested validation selection.

Table 31 is a separate foundation-model audit rather than a new model claim. Pretraining has a positive matched macro-CCC point difference in all four audited protocols, including participant and joint holdout, but the corresponding unseen-music subject probe is also substantially higher for pretrained features. This external source therefore qualifies, rather than reverses, the main case study: pretrained features have higher report-prediction point estimates in this auditory source while retaining participant structure, so neither the positive gain nor a single identity probe supports a general invariant- emotion conclusion.

Table 25: Selected paired macro-CCC comparisons. Intervals are hierarchical paired-bootstrap 95% intervals.

Comparison	Protocol	ΔCCC	Interval	Scope
LaBraM pretrained − random	Stimulus	0.0706	[0.0396, 0.1143]	Known subjects, new stimuli
LaBraM pretrained − random	Subject	0.0015	Includes zero	New subjects
LaBraM pretrained − random	Joint	-0.0008	Includes zero	New subjects and stimuli
Identity overlap − joint, direct EEG	Matched support	0.1511	[0.0349, 0.2156]	Same test and fit counts
Identity overlap − joint, full pipeline	Matched support	0.2111	[0.1101, 0.2706]	Includes prior and residual
Max subject removal − zero	Stimulus	-0.0656	[-0.107, -0.0272]	LaBraM intervention
Subject removal − equal-rank PCA	Stimulus	-0.0364	[-0.0624, -0.0118]	Direction specificity
Subject removal − equal-rank random	Stimulus	-0.0631	[-0.1028, -0.0263]	Direction specificity
Condition prior − EEG-only	External subject	0.3457	[0.304, 0.3884]	New subjects, repeated conditions

Table 26: Semantic-prior component audit. Positive values favor the full prior; intervals use the holdout identity required by each protocol.

Protocol	Omitted component	Full − ablated CCC	95% interval
Stimulus	Nominal category	0.3049	[0.2089, 0.3416]
Subject	Description text	0.0544	[0.0384, 0.071]
Joint	Nominal category	0.4011	[0.2557, 0.4591]
Joint	Context	0.0012	[-0.004, 0.0064]
Joint	Intensity code	-0.01	[-0.0296, 0.0055]
Joint	Description text	-0.0129	[-0.0529, 0.0261]

O Compute and Software Environment

Transparent features, linear baselines, probes, paired statistics, artifact compilation, and tests run in a CPU environment on Linux x86-64 with Python 3.11, NumPy 2.4.6, pandas 3.0.5, SciPy 1.17.1, scikit-learn 1.9.0, and MNE 1.12.1. Foundation feature extraction and EEGNet use one NVIDIA RTX 6000 Ada Generation GPU with 49,140 MiB memory, PyTorch 2.11.0, and CUDA 13.0. CPU-only adversarial residual runs use four threads to avoid competing with GPU jobs.

The typed recomputation graph separates feature extraction, evaluation, statistics, and manuscript compilation. Exact historical wall-clock and energy totals were not captured consistently for every preliminary or failed run; the release therefore reports component environments and DAG boundaries rather than inventing a project-wide total. The final cold-run release log is designed to record per-step wall times and will be published with the executable artifact. This limitation does not affect frozen scores but constrains precise compute-cost replication.

P Release, Licensing, and Reproduction Boundaries

The anonymous package contains code, tests, configurations, evaluation cards, dataset and model cards, Croissant metadata, generated table sources, and claim configuration. It explicitly excludes raw EEG, source feature arrays, foundation checkpoints, movie, music, and image stimuli, office documents, Git history, and machine-local paths.

`metadata` mode validates the benchmark contract without source data. `smoke` uses deterministic synthetic trials to exercise splitting, baselines, probes, and a small registry. `full` requires an explicit user-provided data root and verifies frozen input hashes and paper assets. The typed `recompute`

Table 27: Identity-probe summary. Permuted values use the same representation, split, and estimator with shuffled training labels.

Representation and task	Holdout	Observed	Permuted	Chance
LaBraM pretrained, subject	Stimulus	0.9642	0.0192	0.0333
LaBraM random, subject	Stimulus	0.1733	0.0425	0.0333
DE, dataset	Subject	0.8987	0.3913	0.3333
LaBraM pretrained, stimulus	Subject	0.0234	0.0260	0.0238
External channel, condition	Subject	0.2486	0.1676	0.1667
External global, condition	Subject	0.2134	0.1692	0.1667

Table 28: Complete ds006866 v2 affect audit. Entries are five-seed mean macro-CCC. BP denotes 320-dimensional channel bandpower; Pre. and Rand. denote frozen pretrained and fixed-random LaBraM. Hybrid is the transferable condition-family prior plus the EEG residual.

Protocol	Prior	EEG BP	EEG Pre.	EEG Rand.	Hybrid BP	Hybrid Pre.	Hybrid Rand.
Unseen participant	0.3776	0.0369	0.0335	0.0198	0.3425	0.3742	0.3797
Nonsocial → social, repeated participant	0.3757	0.2319	0.2604	0.2539	0.5144	0.5507	0.5476
Social → nonsocial, repeated participant	0.3780	0.2226	0.2526	0.2455	0.5116	0.5486	0.5462
Nonsocial → social, unseen participant	0.3704	0.0208	0.0323	0.0152	0.3318	0.3718	0.3726
Social → nonsocial, unseen participant	0.3792	0.0172	0.0289	0.0220	0.3472	0.3761	0.3844

`plan/run` interface covers trial construction, feature extraction, evaluation, probes, statistics, and paper compilation; `-resume` skips a step only when all declared outputs exist.

The shortest exact verification sequence from a clean checkout is:

```
conda env create -f environment.yml
conda activate openaffect-eeg
python -m pip install -e ".[smoke]"
python scripts/run_benchmark.py metadata \
  --output-root /tmp/openaffect-metadata
python scripts/run_benchmark.py smoke \
  --output-root /tmp/openaffect-smoke
export D=/path/to/OpenAffect-EEG-data
python scripts/run_benchmark.py full \
  --data-root "$D" --output-root "$D/reproduction"
python scripts/build_neurips_submission.py
```

The first two modes never read EEG. Full mode verifies the frozen registry and rebuilds evidence-linked tables; it is not mislabeled as raw feature extraction or model retraining. The separately documented `recompute_benchmark.py plan/run` commands expose that typed raw-to-paper DAG and require explicit paths to source data, checkpoints, and foundation-model repositories.

Full-mode verification of frozen outputs is not described as a cold download and multi-day retraining run. The existing isolated-environment verification was performed on the same compute host and is not represented as independent laboratory reproduction.

Q Machine-Readable Attachments

The full registry retains per-seed identity results, all paired model comparisons, band and region sensitivity cells, prediction-key audits, source hashes, and unsupported protocol states. These files prevent selective compression of null or unfavorable results into the PDF. Region analyses are predictive sensitivity tests and are not interpreted as neural source localization.

R Position Relative to Closest Evaluation Resources

Table 29: Matched ds006866 v2 identity probes. Each cell is observed/permuted five-seed mean balanced accuracy. Chance reflects the fitted class set.

Protocol	Probe	BP	Pretrained	Random	Chance
Unseen participant	Six-condition	0.2448/0.1669	0.1713/0.1667	0.1675/0.1653	0.1667
Nonsocial → social, repeated participant	Participant	0.9910/0.0066	0.9997/0.0074	0.9963/0.0057	0.0068
Social → nonsocial, repeated participant	Participant	0.9890/0.0063	0.9999/0.0045	0.9973/0.0046	0.0068
Nonsocial → social, unseen participant	Condition family	0.4091/0.3238	0.3399/0.3317	0.3306/0.3333	0.3333
Social → nonsocial, unseen participant	Condition family	0.4206/0.3268	0.3384/0.3336	0.3308/0.3315	0.3333

Table 30: Authorized SEED-V aggregate audit. Values are mean trial macro-F1 across the frozen result rows; Δ is SDSA minus raw temporal. No participant-level or trial-level values are released.

Protocol	Raw temporal	SDSA	Δ	95% interval
Strict participant holdout	0.2418	0.2297	-0.0121	[-0.0439, 0.0178]
Joint participant–stimulus holdout	0.1788	0.1856	0.0068	[-0.0154, 0.0292]

Table 31: Independent MusicEEG frozen LaBraM audit. Values summarize five fixed split seeds; Δ is pretrained minus fixed-random macro-CCC. Subject probes predict training-seen participants under unseen music.

Protocol	Pretrained CCC	Random CCC	Δ CCC
Random trial	0.0923	0.0191	0.0732
Unseen participant	0.0697	-0.0006	0.0703
Unseen music	0.0847	0.0145	0.0703
Unseen participant and music	0.0422	0.0085	0.0337
Unseen-music subject probe (BA)	0.9136	0.4239	—
Permuted subject probe (BA)	0.0313	0.0297	chance 0.0323

Table 32: The closest resources address complementary layers of the problem. The final column states the additional evaluation object introduced here, rather than asserting that prior work is invalid.

Resource	Primary evaluation object	Additional OpenAffect-EEG object
Leakage audit [19]	Trial-segment and test-data leakage on DEAP	Subject, stimulus, joint, context, and dataset identity contracts
mdJPT [41]	Multi-dataset affect pretraining and new-dataset transfer	Deployment-specific identity isolation plus matched random encoders
EMOD [9]	Discrete/continuous label alignment in a common V–A space	Self-report prior/residual separation and stimulus provenance states
EEG-FM-Compass [23]	Cross-task FM comparison under subject transfer and calibration	Affect-specific stimulus and condition audits with predictor-linked interventions
FMScope [22]	Subject-axis diagnosis and erasure across clinical EEG-FMs	Joint subject–stimulus–context contracts and report-level affect estimands
EEG concept audit [37]	Encoded-versus-used physiological concepts under erasure	Experiment-metadata priors separated from trial-level self-reports
Emotion protocol audit [35]	Subject-dependent versus subject-disjoint SEED evaluation	Stimulus, joint, context, and dataset isolation with provenance states
OpenAffect-EEG	Executable evidence and applicability contract	Hash registry, claim ledger, cards, blocked cells, and matched controls

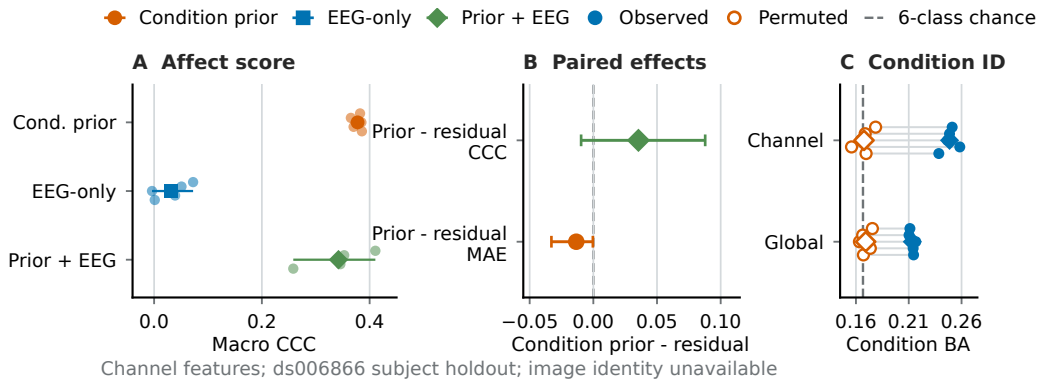


Figure S6: Initial ds006866 subject-holdout condition audit. The condition prior, direct EEG, and condition-plus-EEG residual are reported separately; paired residual effects and condition probes use the five fixed seeds. Image identity is unavailable, so this is condition-level evidence and cannot support a stimulus-holdout claim.

S Historical Overview Figures

The following approved figures retain earlier evidence without changing its source version. The initial data layer covers the original four datasets; Urban is the later source described above. These figures are not the latest matched-increment tables and must not be read as latent-emotion decompositions.

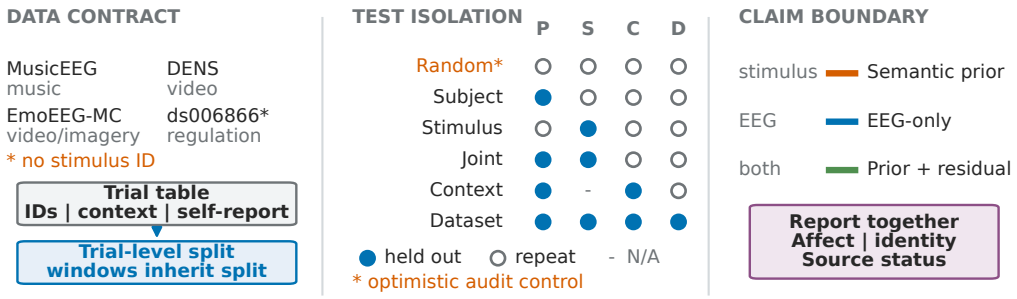


Figure S7: Original trial-first data and evaluation contract. Stimulus identity is unavailable for ds006866. The later Urban source and resource-matched predictor pairs are specified in the main methods and latest-evidence appendix.

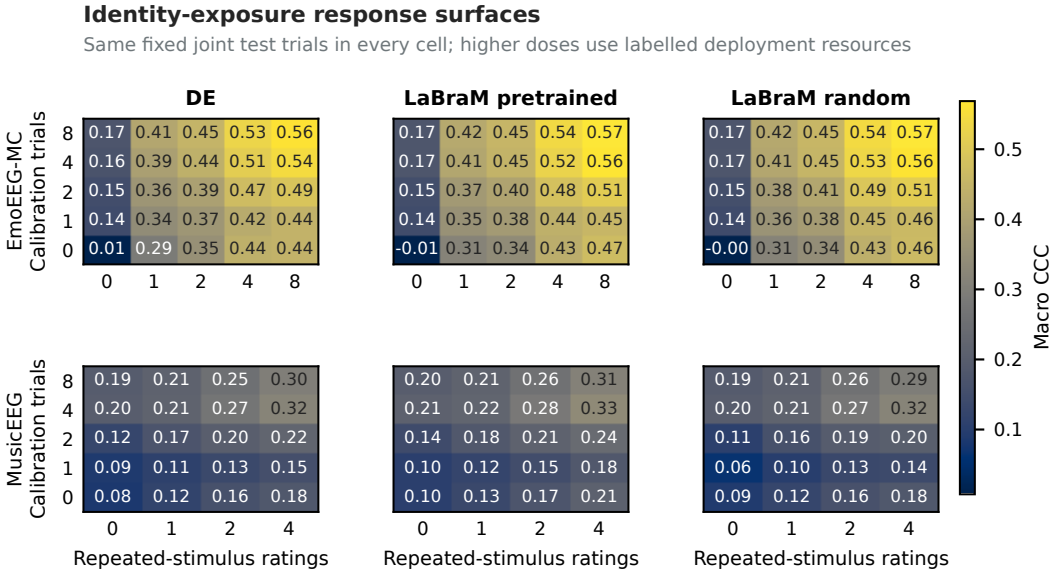


Figure S8: Original EmoEEG-MC and MusicEEG total calibrated prediction surfaces on fixed trials. This is total system CCC, not EEG added value. The new coverage analysis and matched contrasts are reported separately in the main tables; MusicEEG retains its original feasible grid rather than being relabelled as a full maximum-dose grid.

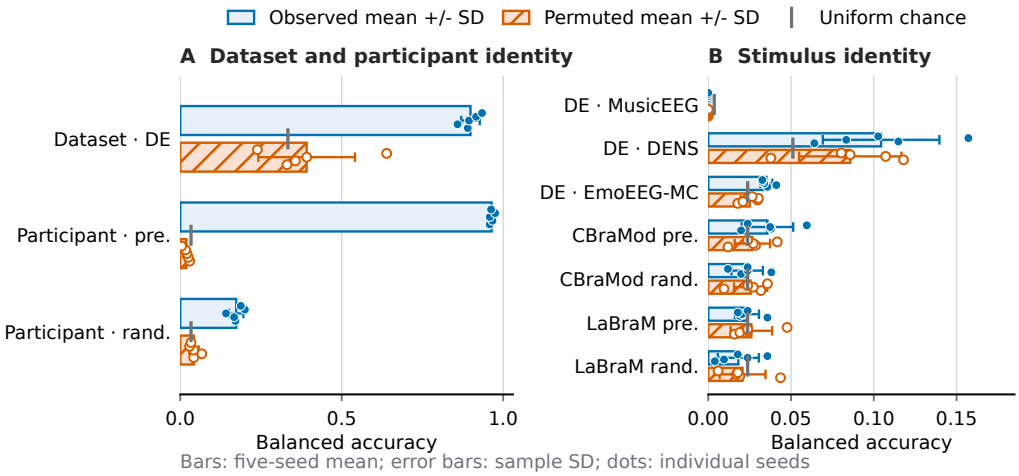


Figure S9: Original representation probes. Bars show the five-seed mean, error bars show sample standard deviation, and points retain every seed. Observed and permuted identity decoding describe available information, not proof that an affect predictor uses identity or that identity is a pure nuisance.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Introduction state the benchmark contributions, identity assumptions, and protocol-dependent scope; the Results and Limitations preserve those same boundaries.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion contains a dedicated Limitations paragraph, with further source-metadata, compute-accounting, and licensing limitations in the appendix.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: Section 3 defines resource-conditioned score contrasts and their arithmetic identity, not a theorem about latent emotion or a causal decomposition.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Sections 3–5 and the v14 appendix specify the fixed support, model selection, resource budgets, five-seed refits, rotation design, joint statistics, and pinned FACED processing. The development appendices retain earlier supports. Released aggregate archives rebuild the tables; separate source-data commands describe full experimental reproduction.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The release provides v14 code, configurations, protocol amendments, aggregate result tables and hashes, and separate aggregate-only and source-data reproduction commands. Source recordings and model weights must be obtained under their providers' terms. The clean-install acceptance is model-executed and author-assisted, not an independent-human or full raw-to-paper reproduction; submission-time anonymous access must match this release.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 4 gives the evaluation logic and model families; The source, protocol, model, and statistics appendices provide split ratios, adaptation details, validation rules, crops, patches, controls, and statistical settings.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The Results and v14 appendix distinguish older fixed-fit intervals from FACED joint participant/stimulus, executed-seed and refit-rotation intervals. FACED equivalence uses pre-outcome margins and a 90% decision interval; EEGNet uncertainty, failed locked controls and exploratory redesigned controls are retained. Older five-block simulations do not validate the v14 bootstrap or cellwise equivalence.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: The compute appendix reports CPU/GPU type, GPU memory, software versions, and component boundaries, but exact wall-clock and energy totals were not logged consistently for every preliminary and failed run.

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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Sections 6–7 discuss scoped evaluation benefits, privacy and identification risk, and why self-report prediction is not clinical or private-state access.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [Yes]

    Justification: The release excludes raw EEG, source derivatives, restricted media, checkpoints, and real-identity linkage; identity probes are documented as risk measurements rather than identification tools.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The source and current-evidence appendices cite source versions and unresolved license caveats, including the Urban snapshot's inconsistent declarations. Model cards record code, checkpoint provenance, and licensing boundaries.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The final anonymous release contains data/model/evaluation cards, valid Croissant metadata, protocol status, limitations, reproducibility modes, and hashes linking the online artifact to this manuscript revision.

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